Parent-of-Origin Rare Variant Gene-Based Burden Tests

Master Thesis in Biostatistics (STA495)

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Abstract

Standard Genome-Wide Association Studies (GWAS) predominantly evaluate additive genetic effects, often operating under the assumption of parental symmetry. However, genomic imprinting—an epigenetic process leading to parent-of-origin-specific monoallelic expression—introduces a layer of biological complexity that traditional models may dilute or fail to detect. This project investigates the parent-of-origin effects (POEs) of rare, high-impact mutations by leveraging Whole Exome Sequencing (WES) data and parent-of-origin-aware gene-based burden tests. By aggregating rare functional variants at the gene level, burden tests provide the better statistical power to detect associations that are otherwise unattainable through single-variant analysis of low-frequency alleles.

Utilizing data from approximately 110,000 individuals in the United Kingdom Biobank (UKB), we employed a surrogate-parent phasing pipeline to resolve maternal and paternal haplotypes. We then performed independent burden associations for each parental haplotype using the REGENIE framework to derive a differential Z-score statistic. Our analysis followed a three-tiered discovery strategy: targeted validation of known signals, a hypothesis-driven screen of 249 known imprinted genes, and an agnostic exome-wide discovery scan across 62 complex traits.

Our results confirm the presence of parent-specific genetic architecture across a broad spectrum of human traits. We successfully validated the approach by replicating established signals and identified multiple associations through our tiered discovery strategies. These findings were predominantly concentrated within metabolic, liver-related, and hormonal phenotypes, revealing distinct patterns of inheritance ranging from bipolar effects to classical parent-of-origin effect where one haplotyope is silenced and the other one is expressed.

Despite these findings, statistical power remains a primary limitation due to the rarity of functional variants and the necessity for balanced haplotype representation. This work provides a proof-of-concept that integrating phasing with burden-testing architectures is an effective strategy for uncovering the missing heritability of complex human traits through exome-wide POE discovery.

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Chapter 1

Introduction

Human beings are remarkably similar at the genetic level, sharing approximately 99.9% of their DNA sequence (Jorde and Wooding, 2004). Despite this high degree of conservation, the remaining 0.1% of genomic variation—amounting to millions of base pair differences—accounts for the vast diversity in measurable traits and susceptibility to disease. These observable characteristics, or phenotypes (P), are classically modeled as the cumulative result of an organism's genotype (G), its environment (E), and the interaction between them:

$$P = G + E + G \times E.$$

To decode which of these small genetic differences are responsible for specific traits, Genome-Wide Association Studies (GWAS) are performed. Over the past two decades, GWAS have fundamentally remodeled the landscape of biomedical research by providing a robust framework to link genetic variations to complex traits and human diseases (Witte, 2010). By scanning the genome for millions of small variations, these studies have pinpointed thousands of loci associated with conditions such as cardiovascular disease, type 2 diabetes, and various cancers without requiring prior knowledge of the underlying biology (Walsh *et al.*, 2023; Centre for Proteomic and Genomic Research (CPGR), 2025). This data-driven lens has offered critical insights into molecular pathways and laid the groundwork for precision medicine through the development of polygenic risk scores, which estimate an individual's genetic predisposition to a specific disease or trait (Manolio, 2013; Traversi *et al.*, 2021).

Standard GWAS typically rely on Linear Mixed Models to evaluate these associations while accounting for confounding factors such as population stratification and relatedness. These models predominantly test for additive genetic effects and use specific genetic positions commonly found in the population for testing. However, rare variants may also be associated with complex traits but their occurrence in the genome is so low that even using large cohorts their association may not be detected in standard GWAS. Protein-altering variants—such as missense and loss-of-function (LoF) mutations—are uncovered by large-scale sequencing efforts such as Whole Exome Sequencing (WES). Because individual rare variants lack the statistical power for single-variant testing, gene-based burden tests are employed to aggregate the cumulative effect of deleterious variants within a single gene.

Despite these successes, standard association models typically operate under the assumption of parental symmetry. This additive framework assumes that the phenotypic effect of an allele is independent of its parental origin and mostly assumes that the causal gene is expressed from the two haplotypes similarly, meaning that each allele has a similar effect on the phenotype. However, a significant layer of biological complexity arises from genomic imprinting—an epigenetic process where one allele is silenced in a parent-of-origin-specific manner, leading to monoallelic expression

(Lawson *et al.*, 2013; Morison *et al.*, 2005). Consequently, these genes exhibit parent-of-origin effects (POEs), where a DNA sequence variant influences the phenotype only when present on the transcriptionally active parental haplotype.

In a GWAS framework, this phenomenon manifests as differential associations depending on the parental source of the variant. The parent-of-origin-aware model reveals the underlying asymmetry: for a heterozygous individual, a variant inherited from the father might show a robust association with the phenotype, while the identical variant inherited from the mother remains phenotypically silent. This would occur in a region of maternal silencing (maternal imprinting, paternal expression), where only the paternally derived allele is functionally active (Figure 1.1). By failing to distinguish between these parental haplotypes, standard models may dilute the true genetic signal or fail to detect associations that are only present in a parent-specific context.

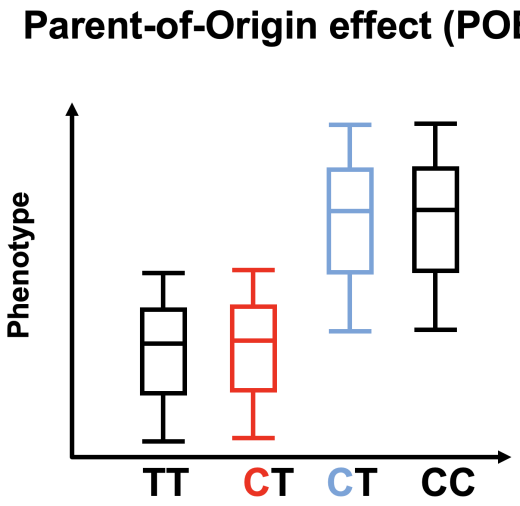


Figure 1.1: Schematic representation of a Parent-of-Origin effect where the maternally inherited alternative allele is silenced.

The theoretical foundation for these effects is rooted in the parental conflict theory (Moore and Haig, 1991). This theory posits an evolutionary divergence between maternal and paternal interests regarding resource allocation during development. Paternally expressed genes generally favor maximal fetal growth to increase individual fitness (Wikipedia contributors, 2025), while maternally expressed genes act to moderate this extraction to preserve maternal resources for future reproductive efforts. This biological "tug-of-war" suggests that at many loci, the maternal and paternal genomes may exert opposing or asymmetric pressures on traits related to growth, metabolism, and energy storage.

While recent work has identified common variant POEs in large biobanks (Lawson *et al.*, 2013; Kong *et al.*, 2018; Hofmeister *et al.*, 2025), the parent-specific impact of rare variant burdens remains largely unexplored du to a lack of: accurate methods to estimate the phase (parental origin) of rare variants (solved by the software SHAPEIT5); large scale cohort with parent-of-origin (solved by Hofmeister *et al.* (2025)); large scale sequencing data (solved by the UK Biobank whole exome sequencing data). This project aims to bridge this gap by investigating the parent-of-origin effects of rare variants. By leveraging WES data and parent-of-origin-aware burden tests, we seek to uncover how the source of rare, high-impact mutations can have an impact on the human phenotypic diversity.

Chapter 2

Methods

2.1 Defintions

2.1.1 Genome-Wide Association Studies

A genome-wide association study (GWAS) is a high-throughput genomic analysis designed to identify statistically significant associations between genetic variants and specific traits or diseases. This approach typically involves performing an independent linear regression at each single nucleotide polymorphism across the genome. In this model, the phenotypic value is treated as the dependent variable, while the genotype serves as the independent variable (Zeng *et al.*, 2015).

2.1.2 Data

This study leverages data from the United Kingdom Biobank (UKB), a comprehensive, long-term prospective cohort study that serves as a premier resource for biomedical research (Bycroft *et al.*, 2018). The UKB provides a robust integration of genome-wide genetic data alongside extensive longitudinal phenotypic and health-related information. Comprising approximately 500,000 participants recruited between 2006 and 2010, the cohort offers statistical power for investigating the complex interplay between genetics, environment, and disease. This vast repository allows for highly powered GWAS, enabling the identification of genetic variants associated with a wide array of complex traits and clinical outcomes. The full list of tested phenotypes and their corresponding phenotype code in the UK Biobank is synthetized in Table 2.1.

2.1.3 Genetic Variants

Genetic variation arises from differences in the DNA sequence at specific genomic loci, characterized as Single Nucleotide Polymorphisms (SNPs). At any given position, the sequence observed in a standard genome assembly is designated as the reference allele, while the variant sequence found within a population is termed the alternative allele. A critical metric is the Minor Allele Frequency (MAF), which represents the frequency of the less common allele in a given population. Variants are typically categorized based on this frequency; for instance, common variants (SNPs) are often defined by a MAF $\geq 1\%$, whereas those falling below this threshold are classified as rare variants (Single Nucleotide Variants (SNVs) Backman *et al.* (2021)).

2.1.4 Gene-Based Burden Test

Given that the Minor Allele Frequency of rare variants is typically too low to support robust single-variant association testing, a gene-based burden approach is employed. This method aggregates multiple genetic variants within a predefined genomic region into a single, continuous burden score, thereby enhancing statistical power by reducing the multiple testing burden and

Table 2.1: List of UKB phenotype codes and their corresponding traits.

UKB	Trait	UKB	Trait
48	Waist circumference	30600	Albumin
49	Hip circumference	30610	Alkaline phosphatase
50	Standing height	30620	Alanine aminotransferase
20015	Sitting height	30630	Apolipoprotein A
20022	Birth weight	30640	Apolipoprotein B
21001	Body mass index (BMI)	30650	Aspartate aminotransferase
22191	Adjusted T/S ratio	30660	Direct bilirubin
23099	Body fat percentage	30670	Urea
23102	Whole body water mass	30680	Calcium
23105	Basal metabolic rate	30690	Cholesterol
23111	Leg fat percentage (right)	30700	Creatinine
23113	Leg fat-free mass (right)	30710	C-reactive protein
23119	Arm fat percentage (right)	30720	Cystatin C
23121	Arm fat-free mass (right)	30730	Gamma glutamyltransferase
23127	Trunk fat percentage	30740	Glucose
23129	Trunk fat-free mass	30750	Glycated haemoglobin (HbA1c)
30000	White blood cell (leukocyte) count	30760	HDL cholesterol
30010	Red blood cell (erythrocyte) count	30770	IGF-1
30080	Platelet count	30780	LDL direct
30090	Platelet crit	30790	Lipoprotein A
30120	Lymphocyte count	30800	Oestradiol
30130	Monocyte count	30810	Phosphate
30140	Neutrophill count	30820	Rheumatoid factor
30150	Eosinophill count	30830	SHBG
30160	Basophill count	30840	Total bilirubin
30170	Nucleated red blood cell count	30850	Testosterone
30190	Monocyte percentage	30860	Total protein
30250	Reticulocyte count	30870	Triglycerides
30500	Microalbumin in urine	30880	Urate
30510	Creatinine (enzymatic) in urine	30890	Vitamin D
30530	Sodium in urine	130708	Type 1 diabetes

capturing cumulative effects. This analysis utilizes Whole Exome Sequencing (WES) data, which specifically targets the protein-coding regions of the genome.
To determine the functional impact of these variants, the WES data is integrated with annotation files that categorize mutations based on their predicted biological consequences. We distinguish between two primary functional categories:

- Predicted loss-of-function mutations (LoF): genetic variations that are predicted to severely reduce or eliminate the normal biological activity of the encoded protein (Rougvie, 2013).
- Missense mutations: Single nucleotide substitutions that result in the replacement of one amino acid with another. These nonsynonymous changes produce an altered protein product that may compromise the protein's structural integrity or biochemical function (Stefl *et al.*, 2013).

The assumption of gene-based burden testing is that variants within a specific functional category share a common direction of effect. This assumption of functional equivalence allows for the aggregation of multiple rare variants into a single, unified burden score, thereby concentrating the statistical signal at the gene level.

2.1.5 Gene Annotations

There is a total of 18'788 protein-coding genes in the 22 autosomes of the human genome for which we have genomic annotations. Autosomes refer to chromsomes that are present in two

copies for every human being, thus excluding sexual chromosomes X and Y. The protein-coding genes refer to exons of the genome, (i.e. genes that are translated into proteins). Gene annotation refers to the addition of biological, functional and clinical meaning to genetic variants.

The gene annotations come from different sources, missense annotations come from alphamissense and LoF annotations come from genebass. We acquired the annotation through a collaboration with researchers from the University of Exeter following the paper from Hawkes *et al.* (2025).

Genetic variants were annotated using Ensembl VEP (v.110), LOFTEE (Karczewski *et al.*, 2020) and UTRannotator (Zhang *et al.*, 2021). They were then assigned to groupings, according to their predicted impact on the coding consequence and location. The different groupings tested were:

- Loss-of-Function (LoF);
- Missense.

Annotations are transcript-specific rather than gene-specific. While genes often produce multiple transcripts with varying structures, these transcripts usually share significant similarities.

2.2 Phasing

2.2.1 Parent-of-Origin Determination with Parents

Analyzing maternal and paternal haplotypes independently necessitates the determination of their parental origin. Given that the human genome is diploid, it is composed of equal autosomal contributions from both parents, the initial objective is to reconstruct the linear sequence of alleles on a single chromosome and assign them to their respective progenitor. This process, known as phasing (or haplotype reconstruction), utilizes statistical frameworks—typically incorporating Hidden Markov Models (HMMs)—to resolve the phase of heterozygous loci and accurately delineate maternal and paternal inheritance (Li and Stephens, 2003; Delaneau *et al.*, 2012; Browning and Browning, 2011).

2.2.2 Phasing with Relatives

Phasing is straightforward when parental haplotypes are accessible; however, within the UK Biobank, parental data is available for only approximately 5,000 participants. This leaves roughly 495,000 individuals whose haplotypes and parental origin cannot be resolved through direct trio (or duo)-based methods. To address this limitation, a method to infer the parental origin of haplotypes by leveraging data from more distant relatives has been developed (Hofmeister *et al.*, 2023). By utilizing this approach, the parental origin of haplotype could be determined for 109,385 individuals within the UK Biobank cohort.
The mechanism of this method relies on identifying a relative within the database through genome-wide genetic similarities. It then assigns the relative to a parental side using chromosome X data for males (males get their chromosome X from their mother) and mtDNA (mitochondrial DNA is transmitted only by the mother) for all individuals. For example, when a male individual shares genetic material with a relative (identity-by-descent (IBD) segments—DNA sequences inherited from a common ancestor (Powell *et al.*, 2010)), this relative is likely to be on the maternal side of the family, since chromosome X is maternally inherited in males. The relative is then referred to as "surrogate mother", as it will be used instead of the actual mother of the individual. The target individual's phased haplotypes are compared to a reference panel composed of these surrogate parental haplotypes alongside haplotypes from 100 unrelated individuals and copies the most similar haplotype segments (as indicated by the black line in

Figure 2.1). We keep track of the reference panel the model is copying from which gives us IBD segments.
Because relatives only share specific segments of the genome with their surrogate parents (and the length of the segments depends on the degree of relatedness, i.e. more distanced relatives will share shorter segments), IBD-based PofO inference initially produces a "patchwork" of labeled segments rather than a continuous chromosome-wide haplotype. To resolve this, a scaffold is constructed. This scaffold acts as a fixed genetic skeleton where the high-confidence PofO-labeled IBD segments serve as "anchors". Finally, all alleles that are not assigned to a parent (grey segments) are phased relatively to the scaffold so that if the phasing is accurate, maternally inherited alleles are phased onto the maternal scaffold which allows us to deduce their parental origin.

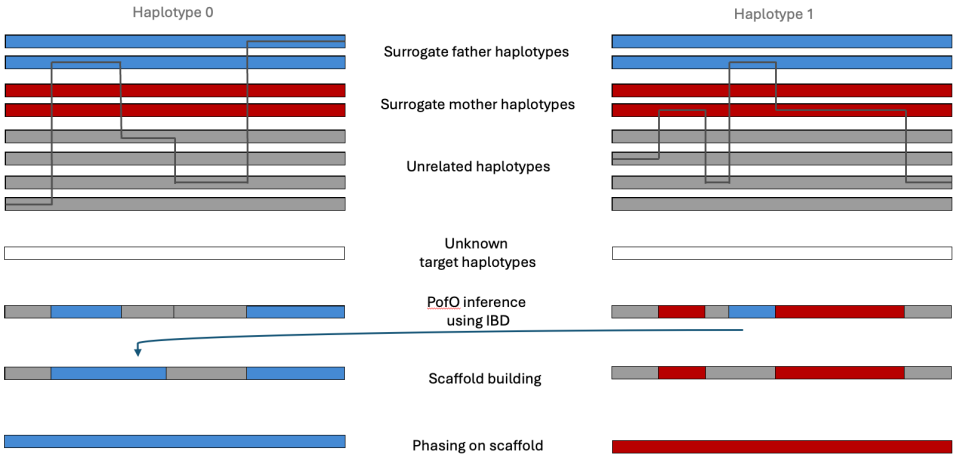


Figure 2.1: Schematic of the surrogate-parent phasing pipeline. The process begins by identifying IBD segments shared between a target individual and surrogate parental haplotypes (top). These PofO-labeled segments are used to build a genetic scaffold (middle). In the final step, the target haplotypes are statistically phased onto this scaffold to produce continuous, chromosome-wide maternal and paternal haplotypes (bottom).

2.3 Analysis

2.3.1 Association Testing

The association analysis is conducted using REGENIE (Mbatchou *et al.*, 2021), a whole genome regression model designed for large-scale genomic datasets. This method requires the integration of high-density genetic data with phenotypic observations. Depending on the data resolution, REGENIE supports multiple testing modalities: Genome-Wide Association Studies (GWAS), which utilize SNP-array data to scan individual variants, and gene-based burden tests, which leverage whole-exome sequencing (WES) data along with functional annotation files to aggregate the effects of rare variants.

The REGENIE framework operates in two distinct stages:

- Step 1: Whole-Genome Model Fitting. In this initial phase, a whole-genome prediction model is fitted to account for complex confounding factors.
  - Level 0: The genome is partitioned into localized blocks, and a ridge regression is applied to each. This regularization technique penalizes individual SNP effects to prevent overfitting while significantly reducing the computational burden associated with high-dimensional genetic matrices. Regenie reads in the M markers in blocks

of B consecutive markers. In each block, a set of ridge regression predictors are calculated for a small range of J shrinkage parameters $\{\tau_1, \ldots, \tau_J\}$

  - Level 1: A cross-validation procedure is employed to generate meta-predictions. These individual-level polygenic scores effectively capture genetic relatedness, population stratification, and polygenic background noise, which serve as offsets in the subsequent step.

- Step 2: Association Testing. The association tests are performed region-by-region, utilizing the predictions from Step 1 (same Step 1 for different Step 2) to maintain statistical power while controlling for inflation.
  - GWAS: An association test is performed for each SNP using a linear regression model.
  - Interaction analyses: gene-gene or gene-environment interactions.
  - Survival analyses.
  - Burden Testing: Rare variants are collapsed within a gene according to functional annotations (masks). A single regression model is then fitted to test the association between the cumulative "burden" of the gene and the phenotype of interest. The multiple variants are aggregated using the following model

$$g(\mu) = w_1 G_1 \beta_1 + \cdots + w_m G_m \beta_m$$

where G_i's represent the single variants included in the test, w_i's and β_i's are weights and effect sizes, respectively and $g(.)$ is is a link function for the phenotypic mean μ.

2.3.2 Parent-of-Origin Burden Tests

Association tests such as burden tests, traditionally identify additive effects, assuming that the phenotype depends on the number of rare variants present in a genetic distance, independent of the parental origin of the DNA segments analyzed. However, some variants are now known to have differential effects depending on whether they were inherited from the father or the mother. In the context of this study, maternal and paternal haplotypes are separated which leads to two independent linear regressions for each gene,

$$Y_{\text{maternal}} = \alpha + \beta_{\text{maternal}} \cdot G_{\text{maternal}} + \epsilon \tag{2.1}$$

$$Y_{\text{paternal}} = \alpha + \beta_{\text{paternal}} \cdot G_{\text{paternal}} + \epsilon \tag{2.2}$$

With Y_{s} being the phenotype explained by the intercept α, the burden score G_{s} with the parental origin s = (maternal, paternal) and an error rate ϵ. β's are the estimates that evaluate the association between the gene burden score and the phenotype assessed.

To assess the difference between paternal and maternal haplotypes derived effect sizes, we use the β's from previous equations (2.1) (2.2) as well as their respective standard errors (SE) to derive a differential Z-score statistic and assess the significance of their difference:

$$Z_{\text{diff}} = \frac{\hat{\beta}_{\text{paternal}} - \hat{\beta}_{\text{maternal}}}{\sqrt{SE^2_{\text{paternal}} + SE^2_{\text{maternal}}}} \tag{2.3}$$

Finally, to transform the differential Z-score to a p-value, we use the standard normal distribution adjusted to a two-tailed test because effect could be in both directions (maternal and paternal), it gives:

$$p = 2 \cdot \Phi(-|Z_{\text{diff}}|) \tag{2.4}$$

In the end, the goal is to determine if the results deviate from the null hypothesis stating that maternal (β_{Mat}) and paternal (β_{Pat}) estimates are equal.

2.3.3 Quality Control

Standard quality control (QC1) was performed to filter variants with $> 10\%$ missingness and variants that significantly deviate from Harry-Weinberg Equilibrium (HWE, p-value $< 1 \times 10^{-300}$). To ensure the validity of the parent-of-origin comparison, a secondary haplotype consistency filter was applied (QC2). We restricted the analysis to variants observed at least once on both the maternal and paternal haplotypes across the cohort. This step prevents the detection of spurious POE signals that could arise from sampling sparsity—where a rare, high-effect variant is present exclusively on one parental haplotype in the dataset. By requiring parental "overlap," we ensure that any significant POE reflects a true biological difference in effect sizes ($\beta_{\text{Mat}} \neq \beta_{\text{Pat}}$) rather than a statistical artifact driven by the absence of data in one of the parental groups.

2.4 Goal

The primary objective of this project is to identify parent-of-origin effects (POEs) within a gene-based burden testing framework. By aggregating rare deleterious variants and assigning them to their respective parental haplotypes, we aim to uncover associations or detect POEs that are otherwise masked by traditional additive models.
To optimize statistical discovery and account for the multiple testing burden, the analysis is structured into three tiers of increasing breadth and decreasing prior probability.

The rationale for this tiered strategy is rooted in statistical power. By performing targeted tests on genes with strong biological priors (tiers 1 and 2), we minimize the number of simultaneous hypotheses tested, thereby lowering the stringency of the multiple-testing correction (e.g., Bonferroni). This increases our capacity to detect signals that may have lower effect sizes but are biologically significant, while the genome-wide scan ensures the potential for novel discovery at a more conservative significance threshold.

2.4.1 Strategy 1: Known Signals

The selection of gene-complex trait pairs is based on the literature. In previous parent-of-origin effects studies (Hofmeister *et al.*, 2025, 2022; Teekakirikul *et al.*, 2021; Benonisdottir *et al.*, 2016; Granot-Hershkovitz *et al.*, 2020; Kim *et al.*, 2021; Kong *et al.*, 2009; Kerin and Marchini, 2020), significant POEs at SNPs loci were identified. Theses SNPs are typically linked to genes in a second step, SNParray data that is used for a GWAS is usually located in non coding regions of the DNA. This strategy uses the genes and traits from POE signals for burden analyses.

2.4.2 Strategy 2: Known Imprinted Genes

There are some genes that are known to be imprinted genes from several previous studies and literature (Jirtle, Randy L., 2026; Genomics England, 2026). The protein coding imprinted genes were selected and the association testing was done on those with all 62 complex traits.

2.4.3 Strategy 3: Genome-wide Discovery

The whole protein coding DNA, (i.e. whole exome, 18,788 genes) was also tested for every complex trait (62).

2.4.4 *TERT* Locus

In addition to the three strategies, we focused on the *TERT* locus and telomere length (Codd *et al.*, 2021) since it was the strongest POE reported by GWAS (Hofmeister *et al.*, 2025).

2.4.5 Bonferroni Correction

To account for the multiple testing burden inherent in high-throughput association studies, a Bonferroni correction was applied to maintain a global family-wise error rate of $\alpha = 0.05$ (Bland, 2015).

First, for the targeted validation of 74 established gene–complex trait pairs (Strategy 1), the significance threshold was set at $p < \frac{0.05}{74} = 6.8e-04$, corresponding to a $-\log_{10}(p)$ value of 3.2.

Second, for the candidate screen of 249 known imprinted genes across 62 phenotypes (Strategy 2), we utilized a corrected threshold of $p < \frac{0.05}{249 \cdot 62} = 3.2e-06$ corresponding to a $-\log_{10}(p)$ value of 5.5.

Third, for the exome-wide discovery scan (Strategy 3), testing 18,788 genes across 62 phenotypes necessitated a highly stringent threshold of $p < \frac{0.05}{18'788 \cdot 62} = 4.3e-08$, which corresponds to a $-\log_{10}(p)$ value of 7.4.

Finally, we also summarized suggestively significant associations for results that did not pass the most stringent p-value correction but still passed the a threshold that was not corrected for the number of phenotypes ($p < \frac{0.05}{18'788} = 2.7e-06$, which corresponds to a $-\log_{10}(p)$ value of 5.6)

Note that given that transcripts within a single a gene are highly correlated, we corrected the number of tests by the number if genes and not by the number of transcripts.

Chapter 3

Results

The results of the parent-of-origin (PofO) association analyses are visualized using Manhattan plots (Figures 3.1, 3.4 and 3.7), in which the x-axis represents the genomic position and the y-axis displays the $-\log_{10}(p)$ transformation of the p-value derived from the differential Z-score statistic (see subsection 2.3.2). This statistic quantifies the magnitude of the difference in effect sizes between maternally and paternally inherited haplotypes (β_{mat} vs. β_{pat}). Accordingly, higher $-\log_{10}(p)$ values reflect stronger evidence against the null hypothesis of equal parental effects, indicating a potential parent-of-origin mechanism underlying the association between the locus and the trait of interest.

The significant associations identified across our three primary strategies are summarized in Table 3.1. Additionally, Table 3.2 contains "suggestive" results—associations that did not pass the most stringent Bonferroni correction for 62 phenotypes but exceed the threshold for a single-trait hypothesis ($p < \frac{0.05}{18'788} \approx 2.7e-06$). These results highlight potential PofO signals that may reach exome-wide significance in larger cohorts.

3.1 Strategy 1: Known Signals

We first targeted 74 gene-phenotype pairs with established PofO associations (see subsection 2.4.1). As shown in the Manhattan plot for this targeted tier (Figure 3.1), the most prominent signal was observed at the ***KCNQ1*** locus in association with standing height ($Z_{\text{Diff}} = -3.44$, $p = 5.81 \times 10^{-4}$), which lands below the Bonferroni corrected significance threshold ($p < 6.76e-04$). This association included variants with MAF $< 0.001\%$. Analysis of rare loss-of-function (LoF) variants revealed a significant bipolar effect: maternal inheritance was associated with an increase in height ($\beta_{mat} = 0.160$), while paternal inheritance was associated with a decrease ($\beta_{pat} = -0.282$; Figure 3.2).

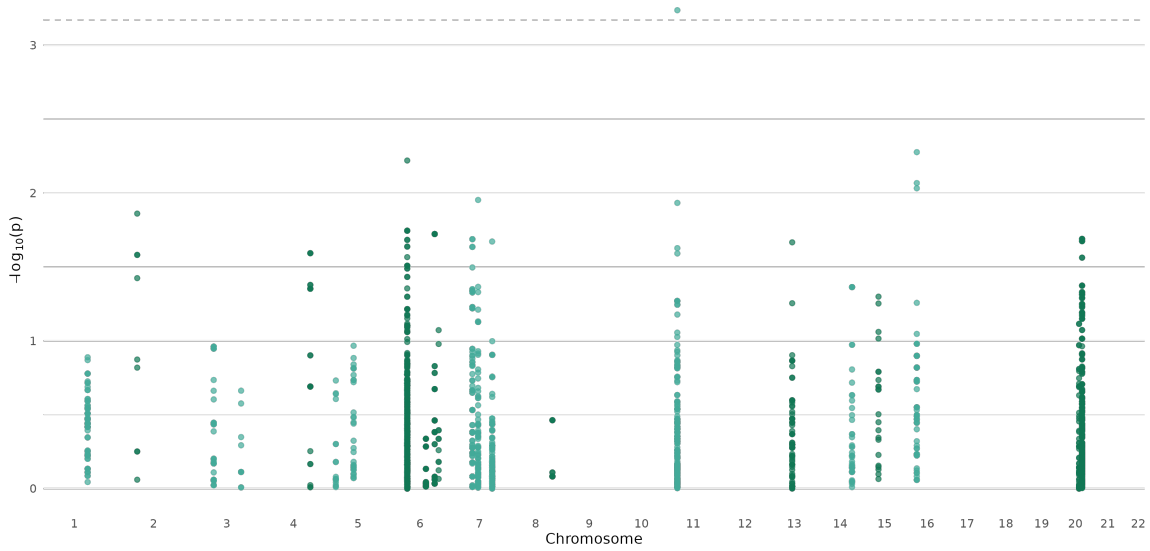


Figure 3.1: Manhattan plot of $-\log_{10}$(differential p-values) (y-axis) alongside the genetic positions (x-axis). Each dot represent a transcript-trait pair tested in strategy 1 (subsection 2.4.1). Light and dark grey indicate odd-numbered and even-numbered chromosomes, respectively. The grey dashed line represents the Bonferroni corrected significance threshold.

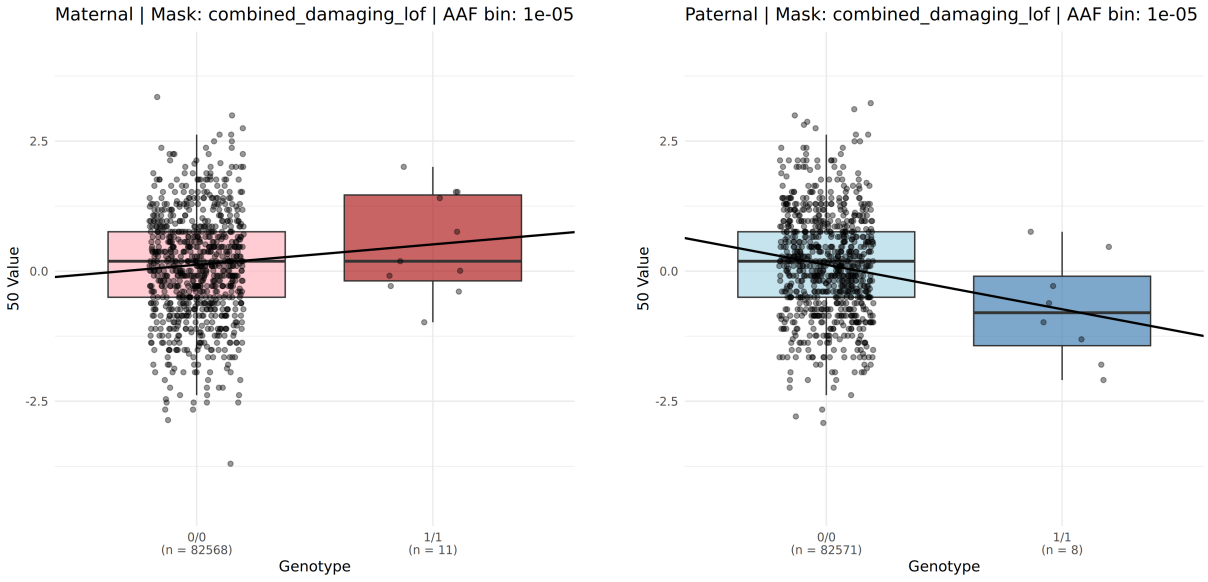


Figure 3.2: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized standing height (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *KCNQ1* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

Figure 3.3 shows a lack of statistical power: the observed $-\log_{10}(p)$ values fall consistently below the expected null distribution (the grey dashed $y = x$ line). This downward deviation indicates that the analysis is underpowered at this tier.

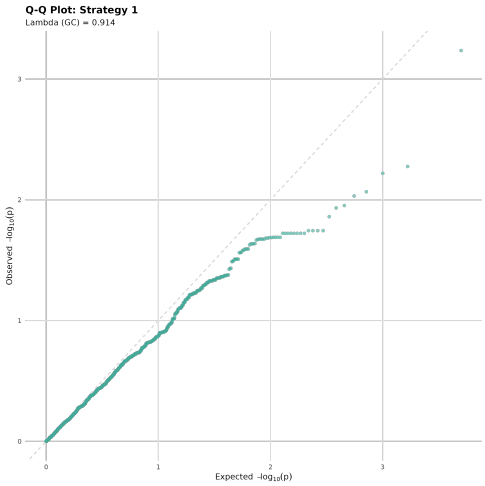


Figure 3.3: QQ plot for Gene-Trait Pairs. The grey line indicates the expected distribution under the null hypothesis.

3.2 Strategy 2: Imprinted Gene Screen

We expanded the analysis to a hypothesis-driven screen of 249 known imprinted genes across 62 phenotypes (Figure 3.4). For this analysis, the significance threshold is at $p < 3.2e-06$. We identified one significant association ($-\log_{10}(p) > 5.5$) and three suggestive ($-\log_{10}(p) > 5$):

- ***SLC4A7***: LoF variants (MAF $< 0.01\%$) showed a significant PofO association with Lipoprotein A levels ($\beta_{\text{Mat}} = -0.59$, $\beta_{\text{Pat}} = 0.62$, $Z_{\text{Diff}} = -4.70$, $p = 2.7 \times 10^{-6}$; Figure A.1), maternally inherited LoF variants in *SLC4A7* are associated with a decrease in lipoprotein A level, whereas the same variants, paternally inherited are associated with an increase of the trait. Note that only 4 individuals carried a LoF variant on the maternal haplotype, and same for the paternal haplotype.

- ***CST1***: Although not significant, there is moderate evidence for a divergence of association between LoF variants (MAF $< 0.01\%$) of *CST1* and Lipoprotein A ($\beta_{\text{Mat}} = -0.55$, $\beta_{\text{Pat}} = 0.60$, $Z_{\text{Diff}} = 4.56$, $p = 5.2 \times 10^{-6}$; Figure A.3). Maternally inherited variants are associated with a decrease in lipoprotein A level and paternally inherited variants are associated with an increase in the trait.

- ***NAPRT***: Although not significant as well, this PofO difference of association between missense variants (MAF $< 0.1\%$) on *NAPRT* and Apolipoprotein B levels ($\beta_{\text{Mat}} = -0.28$, $\beta_{\text{Pat}} = 0.18$, $Z_{\text{Diff}} = 4.51$, $p = 6.4 \times 10^{-6}$; Figure A.2) still shows a suggestive association. Paternally inherited missense variants lead to a increase in phenotype, as opposed to maternally inherited missense that lead to a decrease in apolipoprotein B level.

- ***PLG***: At this locus, the difference in association for missense variants (MAF $< 0.1\%$) considering their parental origin gives a suggestive result. Maternally inherited missense variant are associated with a decreased erythrocyte count whereas paternally inherited missense variants are associated with an increased erythrocyte count ($\beta_{\text{Mat}} = -0.14$, $\beta_{\text{Pat}} = 0.15$, $Z_{\text{Diff}} = 4.45$, $p = 8.53 \times 10^{-6}$; Figure A.4).

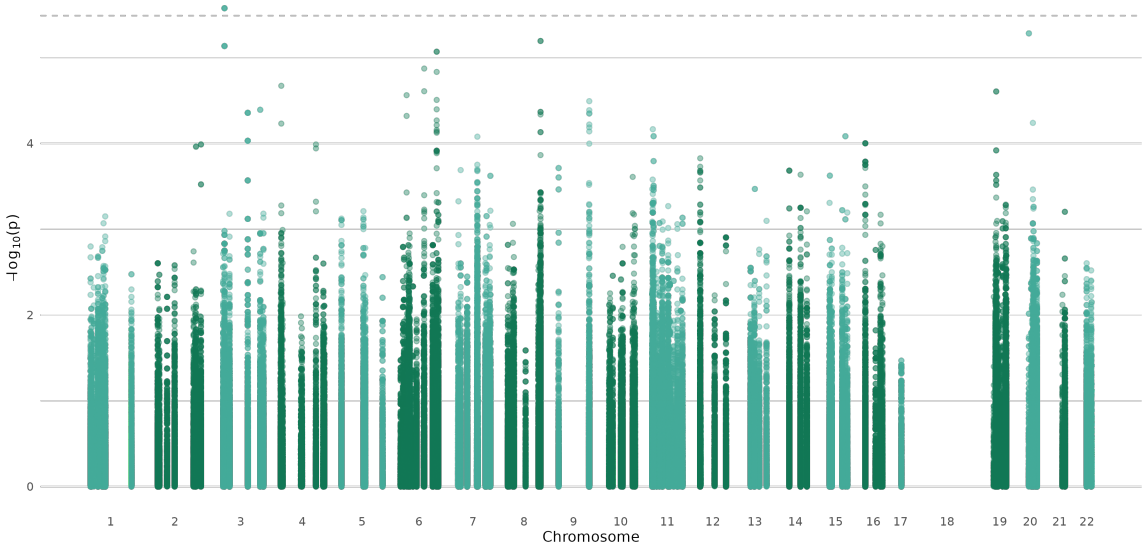


Figure 3.4: Manhattan plot of $-\log_{10}$(differential p-values) (y-axis) alongside the genetic positions (x-axis). Each dot represent a transcript-trait pair tested in strategy 2 (subsection 2.4.2). Light and dark grey indicate odd-numbered and even-numbered chromosomes, respectively. The grey dashed line represents the Bonferroni corrected significance threshold.

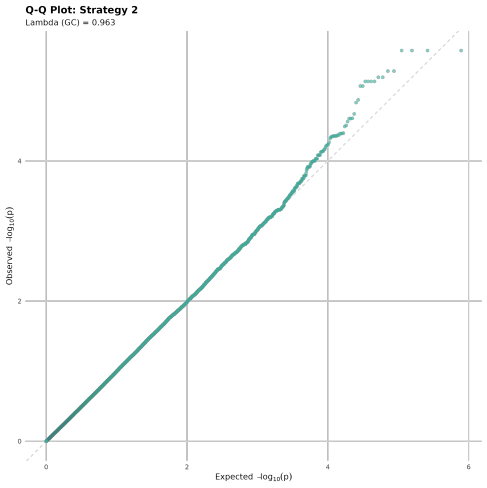


Figure 3.5: QQ plot for Imprinted Genes.

Figure 3.5 exhibits good model calibration, while the terminal upward deviation confirms the detection of parent-of-origin signals within the candidate imprinted gene set.

Table 3.1: Significant association results from Figures 3.1, 3.4, 3.7. Annotation column refers to the variant assessed; AAF bin (alternative allele frequency bin) refers to the frequency of variants assesed; N's, β's and SE's are computed using REGENIE (Mbatchou *et al.*, 2021); $P_{\text{Differential}}$ is computed through the differential Z-score (see section 2.3); trait refers to the phenotype tested; strategy refers to the prior used for significance threshold (see section 2.4).

Chr	Gene	Annotation	AAF Bin	N_{Mat}	β_{Mat}	SE_{Mat}	N_{Pat}	β_{Pat}	SE_{Pat}	$P_{\text{Differential}}$	Trait	Strategy
1	PATJ	Missense	all	81760	0.106	0.015	73176	0.001	0.005	1.4e-11	Apolipoprotein B	3
1	PATJ	Missense	all	82159	0.110	0.015	73539	0.003	0.005	1.2e-11	Cholesterol	3
1	PATJ	Missense	all	82029	0.109	0.015	73419	0.002	0.005	1.5e-11	LDL	3
1	LRRC8C	Missense	all	13710	-0.001	0.004	80006	0.114	0.019	3.0e-09	Alkaline phosphatase	3
1	ADAR	Missense	all	56214	0.003	0.004	80255	0.097	0.016	1.1e-08	Cholesterol	3
2	AAK1	Missense	all	80621	0.089	0.014	38517	0.003	0.003	8.7e-10	Alkaline phosphatase	3
2	AAK1	Missense	all	80572	0.101	0.014	38492	-0.002	0.003	1.7e-12	Alanine aminotransferase	3
2	AAK1	Missense	all	80214	0.118	0.015	38342	-0.001	0.003	4.1e-15	Apolipoprotein B	3
2	AAK1	Missense	all	80300	0.092	0.014	38362	-0.001	0.003	3.1e-10	Aspartate aminotransferase	3
2	AAK1	Missense	all	80611	0.133	0.015	38517	0.000	0.003	3.2e-18	Cholesterol	3
2	AAK1	Missense	all	80571	0.077	0.013	38496	0.001	0.003	4.0e-08	Gamma glutamyltransferase	3
2	AAK1	Missense	all	80481	0.116	0.015	38446	0.000	0.003	5.6e-14	LDL direct	3
2	COL3A1	Missense	all	21741	-0.002	0.003	80289	0.094	0.015	1.3e-09	Alkaline phosphatase	3
2	COL3A1	Missense	all	21732	-0.003	0.004	80241	0.091	0.016	1.3e-08	Alanine aminotransferase	3
2	COL3A1	Missense	all	21632	-0.005	0.004	79887	0.126	0.017	2.4e-14	Apolipoprotein B	3
2	COL3A1	Missense	all	21733	-0.006	0.004	80279	0.145	0.017	1.6e-18	Cholesterol	3
2	COL3A1	Missense	all	21700	-0.005	0.004	80152	0.131	0.017	5.8e-15	LDL direct	3
3	**SLC4A7**	**LoF**	**0.0001**	**4**	**-0.586**	**0.181**	**4**	**0.619**	**0.181**	**2.7e-06**	**Lipoprotein A**	**2**
3	SCN10A	Missense	all	52695	-0.002	0.003	80740	0.092	0.013	5.7e-12	Apolipoprotein B	3
3	SCN10A	Missense	all	52957	-0.002	0.003	81136	0.109	0.014	1.5e-15	Cholesterol	3
3	SCN10A	Missense	all	52875	-0.004	0.003	81007	0.098	0.014	5.5e-13	LDL direct	3
4	CPZ	Missense	all	80835	0.086	0.012	57882	0.006	0.003	1.7e-10	Alkaline phosphatase	3
4	CPZ	Missense	all	80787	0.082	0.013	57849	0.003	0.003	3.3e-09	Alanine aminotransferase	3
4	CPZ	Missense	all	80427	0.115	0.013	57608	0.006	0.004	1.2e-15	Apolipoprotein B	3
4	CPZ	Missense	all	80823	0.130	0.013	57873	0.005	0.004	9.8e-20	Cholesterol	3
4	CPZ	Missense	all	80379	-0.072	0.012	57560	0.000	0.003	2.8e-08	IGF-1	3
4	CPZ	Missense	all	80693	0.120	0.013	57783	0.005	0.004	8.5e-17	LDL direct	3
4	PPEF2	Missense	all	80185	0.101	0.017	17554	0.000	0.004	8.2e-09	Alkaline phosphatase	3
4	PPEF2	Missense	all	79785	0.109	0.018	17472	0.001	0.004	8.2e-09	Apolipoprotein B	3
4	PPEF2	Missense	all	80174	0.141	0.019	17555	0.001	0.004	1.1e-13	Cholesterol	3
4	PPEF2	Missense	all	80048	0.123	0.019	17516	0.000	0.004	1.1e-10	LDL direct	3
6	MUCL3	Missense	all	81108	0.103	0.015	46555	-0.005	0.003	4.9e-13	Alanine aminotransferase	3
6	MUCL3	Missense	all	80752	0.095	0.015	46344	-0.001	0.003	5.6e-10	Apolipoprotein B	3
6	MUCL3	Missense	all	80833	0.079	0.015	46394	-0.007	0.003	1.3e-08	Aspartate aminotransferase	3
6	MUCL3	Missense	all	81146	0.109	0.015	46578	-0.002	0.003	9.2e-13	Cholesterol	3
6	MUCL3	Missense	all	81017	0.106	0.015	46504	0.000	0.003	1.4e-11	LDL direct	3
6	**PLG**	**Missense**	**0.001**	**75**	**-0.135**	**0.046**	**80**	**0.149**	**0.044**	**8.5e-06**	**Erythrocyte count**	**2**
8	**NAPRT**	**Missense**	**0.001**	**37**	**-0.281**	**0.077**	**50**	**0.175**	**0.066**	**6.4e-06**	**Apolipoprotein B**	**2**

9	CRB2	Missense	all	44297	0.005	0.003	80569	0.093	0.015	4.1e-09	Apolipoprotein B	3
9	CRB2	Missense	all	44518	0.005	0.003	80965	0.107	0.015	1.5e-11	Cholesterol	3
9	CRB2	Missense	all	44447	0.005	0.003	80835	0.103	0.015	1.4e-10	LDL direct	3
9	CRB2	Missense	all	40623	0.002	0.003	73789	-0.074	0.013	4.1e-08	SHBG	3
10	WDFY4	Missense	all	70031	-0.001	0.005	81721	0.093	0.015	7.3e-09	Apolipoprotein B	3
10	WDFY4	Missense	all	70378	0.002	0.005	82122	0.120	0.016	3.4e-13	Cholesterol	3
10	WDFY4	Missense	all	70259	0.001	0.005	81991	0.107	0.016	8.2e-11	LDL direct	3
11	**KCNQ1**	**LoF**	**0.00001**	**12**	**0.160**	**0.081**	**8**	**-0.282**	**0.100**	**5.8e-04**	**Standing Height**	**1**
11	OR5W2	Missense	all	24673	-0.007	0.004	80137	0.088	0.016	1.6e-08	Apolipoprotein B	3
11	OR5W2	Missense	all	24795	-0.003	0.004	80533	0.106	0.017	1.2e-10	Cholesterol	3
11	OR5W2	Missense	all	24758	-0.005	0.004	80404	0.098	0.017	1.5e-09	LDL direct	3
11	OR8H3	Missense	all	25746	-0.008	0.003	80176	0.104	0.016	3.0e-12	Apolipoprotein B	3
11	OR8H3	Missense	all	25878	-0.005	0.004	80571	0.119	0.016	1.8e-14	Cholesterol	3
11	OR8H3	Missense	all	25837	-0.006	0.004	80442	0.114	0.016	2.4e-13	LDL direct	3
13	COG3	Missense	all	80179	0.111	0.017	60918	0.004	0.004	4.8e-10	Alanine aminotransferase	3
13	COG3	Missense	all	79824	0.140	0.017	60657	0.003	0.004	5.1e-15	Apolipoprotein B	3
13	COG3	Missense	all	79910	0.106	0.017	60720	0.007	0.004	7.9e-09	Aspartate aminotransferase	3
13	COG3	Missense	all	80216	0.161	0.017	60941	0.003	0.004	6.4e-19	Cholesterol	3
13	COG3	Missense	all	80089	0.146	0.017	60839	0.002	0.004	8.9e-16	LDL direct	3
16	QPRT	Missense	all	16648	-0.008	0.004	79682	0.111	0.019	4.0e-10	Apolipoprotein B	3
16	QPRT	Missense	all	16728	-0.007	0.004	80073	0.113	0.019	3.9e-10	Cholesterol	3
16	QPRT	Missense	all	16705	-0.007	0.004	79946	0.110	0.019	1.7e-09	LDL direct	3
19	ZNF607	Missense	all	80681	0.110	0.013	67156	0.012	0.004	2.4e-12	Alkaline phosphatase	3
19	ZNF607	Missense	all	80276	0.107	0.015	66814	0.008	0.004	1.9e-10	Apolipoprotein B	3
19	ZNF607	Missense	all	80669	0.115	0.015	67148	0.005	0.004	8.2e-13	Cholesterol	3
19	ZNF607	Missense	all	80543	0.111	0.015	67048	0.006	0.004	2.1e-11	LDL direct	3
20	**CST1**	**LoF**	**0.0001**	**5**	**-0.549**	**0.162**	**4**	**0.559**	**0.181**	**5.2e-06**	**Lipoprotein A**	**2**

Table 3.2: Suggestively significant associations from Figure 3.7. Annotation column refers to the variant assessed; AAF bin (alternative allele frequency bin) refers to the frequency of variants assesed; N's, β's and SE's are computed using REGENIE (Mbatchou *et al.*, 2021); $P_{\text{Differential}}$ is computed through the differential Z-score (see section 2.3); trait refers to the phenotype tested; strategy refers to the prior used for significance threshold (see section 2.4).

Chr	Gene	Annotation	AAF Bin	N_{Mat}	β_{Mat}	SE_{Mat}	N_{Pat}	β_{Pat}	SE_{Pat}	$P_{\text{Differential}}$	Trait	Strategy
1	MIB2	Missense	0.001	807	0.030	0.016	754	-0.084	0.017	1.3e-06	Lymphocyte count	3
1	CTH	LoF	0.00001	3	-0.200	0.161	3	0.916	0.161	1.0e-06	Arm fat-free mass	3
2	SLC4A3	Missense	0.0001	309	0.084	0.025	288	-0.094	0.026	8.1e-07	Glycated haemoglobin (HbA1c)	3
3	SLC4A7	LoF	0.0001	4	-0.586	0.181	4	0.619	0.181	2.7e-06	Lipoprotein A	3
3	CHCHD6	Missense	0.0001	55	-0.297	0.065	86	0.105	0.052	1.1e-06	Calcium level	3
5	LIFR	Missense	0.01	1309	-0.043	0.012	1264	0.037	0.012	2.1e-06	Gamma glutamyltransferase	3
5	FYB1	LoF	0.00001	5	0.695	0.213	9	-0.615	0.159	7.8e-07	Calcium level	3
5	SPATA31J1	Missense	0.00001	12	-0.210	0.083	12	0.345	0.083	2.4e-06	Leg fat-free mass	3
6	CNR1	LoF	0.00001	3	-0.919	0.191	4	0.319	0.166	9.9e-07	Sitting height	3
6	MAP7	LoF	0.0001	4	-0.749	0.166	4	0.356	0.166	2.3e-06	Sitting height	3
7	ELFN1	LoF	0.00001	10	-0.523	0.147	8	0.562	0.165	9.0e-07	Apolipoprotein B	3
7	ELFN1	LoF	0.00001	10	-0.496	0.150	8	0.628	0.168	6.2e-07	LDL direct	3
7	HIBADH	LoF	0.0001	3	1.298	0.235	8	-0.133	0.144	2.2e-07	Cystatin C level	3
7	PPP1R3A	LoF	0.0001	7	-0.519	0.107	5	0.303	0.126	6.6e-07	Standing height	3
8	RBM12B	Missense	0.00001	4	-0.658	0.187	7	0.466	0.142	1.8e-06	Creatinine	3
10	SLC25A16	LoF	0.00001	5	-0.823	0.203	4	0.820	0.227	7.3e-08	Alanine aminotransferase	3
10	DHX32	LoF	0.0001	3	0.889	0.165	8	-0.136	0.101	1.2e-07	Leg fat pct.	3
11	CHID1	Missense	0.00001	9	0.499	0.135	8	-0.484	0.143	6.0e-07	SHBG	3
11	OR5J2	Missense	0.01	1016	-0.028	0.010	1008	0.041	0.010	2.5e-06	Sitting height	3
12	ANO2	LoF	0.00001	20	-0.236	0.091	17	0.408	0.099	1.6e-06	Cystatin C level	3
12	C12orf76	Missense	0.00001	4	-0.516	0.150	3	0.561	0.173	2.4e-06	Testosterone	3
14	CTSG	Missense	0.00001	13	0.124	0.113	5	-0.919	0.182	1.0e-06	Cystatin C level	3
14	NRXN3	Missense	0.01	709	-0.041	0.014	686	0.051	0.014	2.6e-06	Lipoprotein A	3
15	EXD1	LoF	0.00001	4	0.483	0.164	8	-0.469	0.116	2.2e-06	Sitting height	3
15	SLTM	LoF	0.0001	10	0.215	0.108	3	-0.850	0.198	2.3e-06	Arm fat pct.	3
15	THAP10	Missense	0.00001	8	0.194	0.103	5	-0.604	0.130	1.5e-06	Basal metabolic rate	3
15	THAP10	Missense	0.00001	8	0.149	0.101	5	-0.640	0.128	1.4e-06	Leg fat-free mass	3
16	TSC2	Missense	0.0001	68	-0.094	0.047	46	0.257	0.057	2.3e-06	Total bilirubin	3
16	NUP93	LoF	0.00001	4	0.360	0.149	4	-0.658	0.149	1.5e-06	Testosterone	3
18	NAPG	LoF	0.00001	5	-0.431	0.134	5	0.459	0.134	2.5e-06	Testosterone	3
19	PLPPR3	LoF	0.001	54	-0.132	0.041	35	0.177	0.051	1.9e-06	Testosterone	3
19	SAMD1	Missense	0.00001	14	0.566	0.115	16	-0.211	0.108	8.3e-07	Platelet count	3
19	FCGBP	LoF	0.00001	3	-0.998	0.198	8	0.158	0.121	6.0e-07	Arm fat pct.	3
19	PIH1D1	Missense	0.00001	9	-0.396	0.109	16	0.256	0.082	1.9e-06	Sitting height	3
19	ZNF320	LoF	0.00001	4	-0.796	0.188	4	0.453	0.187	2.5e-06	Creatinine	3
20	CASS4	LoF	0.0001	9	0.692	0.148	18	-0.169	0.105	2.1e-06	IGF-1	3
20	CASS4	Missense	0.0001	8	0.774	0.157	14	-0.200	0.119	7.7e-07	IGF-1	3
22	APOBEC3D	Missense	0.01	709	0.051	0.017	645	-0.071	0.018	1.3e-06	Total protein	3

3.3 Strategy 3: Exome-wide Discovery

The analysis was once again expanded, this time to an exome-wide scan of 18'788 genes for a divergence of association between parental inherited variants with 62 traits (Figure 3.7). All results in this section included common variants (AAF bin: all, in Table 3.1), meaning that the scan was made for all variant frequencies (not restricted to rare variants with AAF $\leq 1\%$). All frequencies of variants were tested here, however, only common variants (AAF $> 10\%$) showed significance. Several genes showed differences in association with traits depending on parental variant inheritance (only citing the most significant results):

- ***CPZ***: Missense variants showed significant PofO associations with several traits (cholesterol, alkaline phosphatase, alanine aminotransferase, apolipoprotein B, IGF-1 and LDL direct; respective differential values: $Z_{\text{Diff}} = -9.09, -6.39, -5.92, -8.00, 5.56, -8.32$; all p-values below the corrected Bonferroni threshold of 4.3e-08). All signals, except for the association with IGF-1 have the same direction: maternally inherited missense variants are associated with an increase in the respective traits, whereas paternally inherited missense variants do not seem to be associated with a change in the respective phenotype level (Figure A.5). For the IGF-1 trait, maternally inherited missense variants are associated with a decrease in the phenotype and paternally inherited missense variants are not associated with a change in phenotype level.

- ***COG3***: Similarly, we identified significant PofO singals with several phenotype (cholesterol, asparate aminotransferase, alanine aminotransferase, apolipoprotein B and LDL direct) levels based on missense variants at the *COG3* locus. All effects are in the same direction, maternally inherited missense variants are associated with an increase in the respective phenotype and paternally inherited missense variants are not associated with a change of phenotype level.

All results are available in Table 3.1, and plots in section A.1

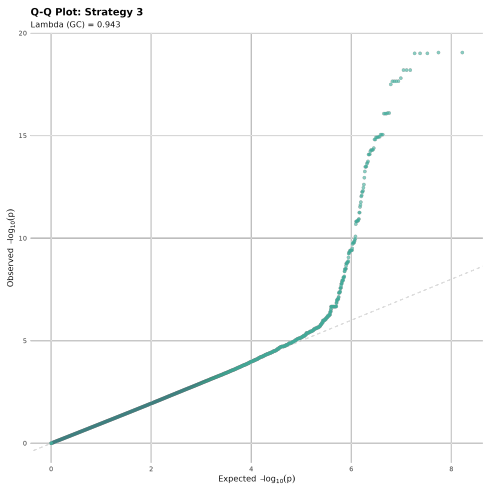


Figure 3.6: QQ plot for the Exome-Wide scan. The grey line indicates the expected distribution under the null hypothesis.

Figure 3.6 demonstrates robust model calibration, featuring a significant upward deviation at the tail that suggests the detection of parent-of-origin associations.

3.4 Exome-Wide Suggestive Associations

In Table 3.2, are summarized suggestive results, that did not pass the significant threshold but nevertheless showed interesting PofO signals. These associations land between the dashed and dotted grey lines on Figure 3.7. Only rare variants (MAF $\leq 1\%$) were included in these suggestive results.

- ***FYB1***: The presence of LoF variants (MAF $< 0.001\%$) on the maternal haplotype is associated with an increased level in calcium compared to the presence of LoF variants on the paternal haploypte, which is associated with a decrease in calcium level ($\beta_{\text{Mat}} = 0.70$, $\beta_{\text{Pat}} = -0.62$, $Z_{\text{Diff}} = -4.94$, $p = 7.8 \times 10^{-7}$; Figure A.6).

- ***CTSG***: Another suggestive PofO association between missense variants (MAF $< 0.001\%$) in the *CTSG* locus with cystatin C level. Maternally inherited missense variants are associated with an increased cystatin C level and paternally inherited variants are associated with an increased phenotype level ($\beta_{\text{Mat}} = 0.12$, $\beta_{\text{Pat}} = 0.92$, $Z_{\text{Diff}} = -4.88$, $p = 1.0 \times 10^{-6}$; Figure A.7).

- ***THAP10***: Basal metabolic rate and leg fat-free mass show suggestive PofO in association for missense ((MAF $< 0.001\%$)) variants. The associations with both phenotypes result in similar estimation in the same directions ($\beta_{\text{Mat}} = 0.19$, $\beta_{\text{Pat}} = -0.60$, $Z_{\text{Diff}} = -4.81$, $p = 1.5 \times 10^{-6}$; respectively $\beta_{\text{Mat}} = 0.15$, $\beta_{\text{Pat}} = -0.64$, $Z_{\text{Diff}} = -4.83$, $p = 1.4 \times 10^{-6}$, for basal metabolic rate and leg fat-free mass; Figure A.8). Maternally inherited missense variants are associated with higher phenotpic values whereas paternally inherited missense variants are associated with lower phenotypic values compared to individuals with no variants.

- ***NAPG*** and ***PLPPR3***: At these two loci, similar PofO signals for LoF variants (MAF $< 0.001\%$ for *NAPG* and MAF $< 0.1\%$ for *PLPPR3*) were identified when testing for association with testoterone. For both genes, loss-of-function variants when maternally inherited are associated to lower testosterone levels as opposed to when paternally inherited LoF which are associated to increased levels of testosterone ($\beta_{\text{Mat}} = -0.43$, $\beta_{\text{Pat}} = 0.46$, $Z_{\text{Diff}} = 4.71$, $p = 2.5 \times 10^{-6}$ for *NAPG* and $\beta_{\text{Mat}} = -0.13$, $\beta_{\text{Pat}} = 0.18$, $Z_{\text{Diff}} = 4.76$, $p = 1.9 \times 10^{-6}$ for *PLPPR3*; Figures A.9 and A.10).

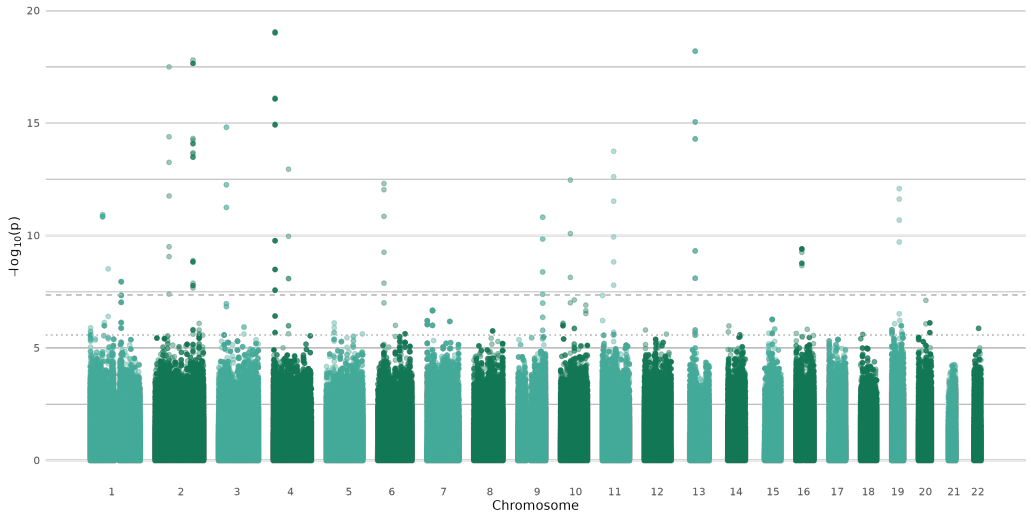


Figure 3.7: Manhattan plot of $-\log_{10}$(differential p-values) (y-axis) alongside the genetic positions (x-axis). Each dot represent a transcript-trait pair tested in strategy 3 (subsection 2.4.3). Light and dark grey indicate odd-numbered and even-numbered chromosomes, respectively. The grey dashed line represents the Bonferroni corrected significance threshold. The grey dotted line represents the threshold for suggestive results (significant threshold corrected by the number of genes and not by the number of genes × number of phenotypes)

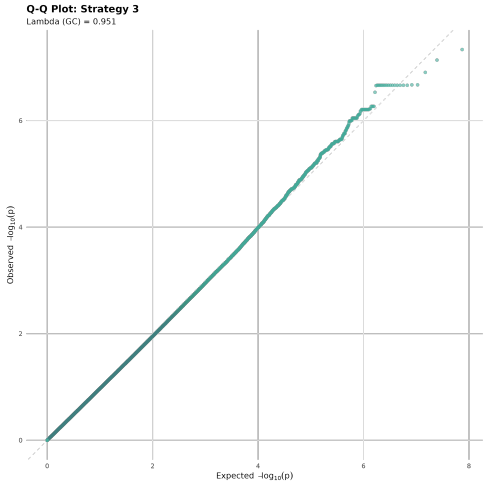


Figure 3.8: QQ plot for the Exome-Wide scan restricted to small MAF variants (MAF $\leq 1\%$). The grey line indicates the expected distribution under the null hypothesis.

The rare-variant restricted QQ plot shown in Figure3.8 remains well-calibrated but shows an attenuated tail lift compared to the full scan, reflecting the expected decrease in discovery power when filtering for low-frequency alleles.

3.5 TERT Analysis

All parent-of-origin effects from Hofmeister *et al.* (2025) were analyzed in strategy 1. We were particularly interested by the TERT locus, since it is the strongest GWAS signal in the paper. However, nothing significant was identified. We explored the possibility that the gene was mis-annotated, as only genomic proximity was used to link the GWAS SNP to the TERT gene. We thus decided to evaluate the association between all genes within a 500 kb window of the GWAS SNP and telomere length, to try to find a better candidate gene than TERT.

For this analysis, we use a p-value threshold of $\frac{0.05}{5} = 0.01$. Nothing was significant; the top gene in the window was GMDS (differential p-value $= 0.052$)

3.6 QC1 and QC2 Comparison

A comparison between p-values associated to differential Z-scores is shown in Figure 3.9 for data out of quality control 1 and data out of quality control 2. Results having the have the same value after QC1 and QC2 (i.e. $y = x$) suggest that the data passed both quality controls without having to rule out low quality data (subsection 2.3.3). Most of the resulting p-values are on or close to the diagonale line for which $y = x$. Some associations clearly deviate from this line, mostly towards QC1, resulting in a highly significant p-value from quality control 1 but very low significance from quality control 2.

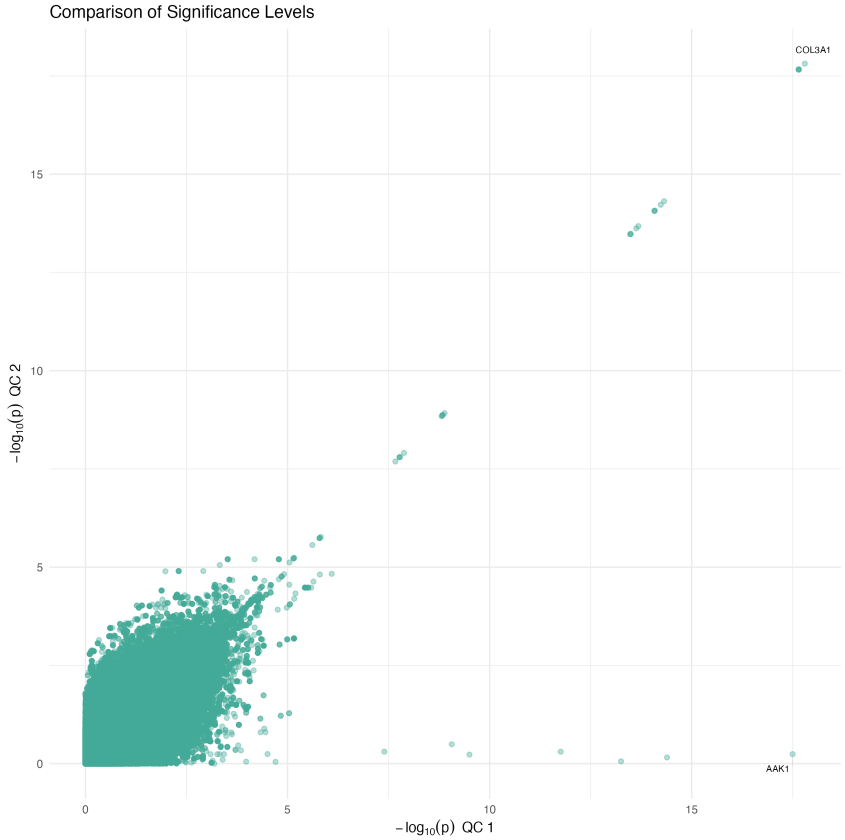


Figure 3.9: QC1 and QC2 comparison plot of $-\log_{10}(p)$-values associated to differential Z-scores from exome-wide gene (chromosome 2) associations with 62 phenotypes.

3.7 GWAS Scan of Significant Signals

Parent-of-origin signals were further analyzed to assess whether the signal of the significant association was driven by the burden score as a unit or to determine of one variant in particular was more strongly associated and potentially responsible for the PofO signal detected. Figure 3.10 shows the $-\log_{10}(p)$ values associated to the differential Z-scores as before but this time the Z-score is not derived from β burden scores from SNPs directly. In the left panel of Figure 3.10 is displayed an example where a particular variant stands out compared to the others. On the left panel is displayed an example of a GWAS where no variant specifically stands out. It could be that the isolated variant in the right panel carries a significant amount of signal for the burden score, to test this, a burden association test could be performed again conditionning on this

specific variant to make sure that the other variants also carry signal and that the burden test is the proper way of testing this or if standard GWAS would be more appropriate.

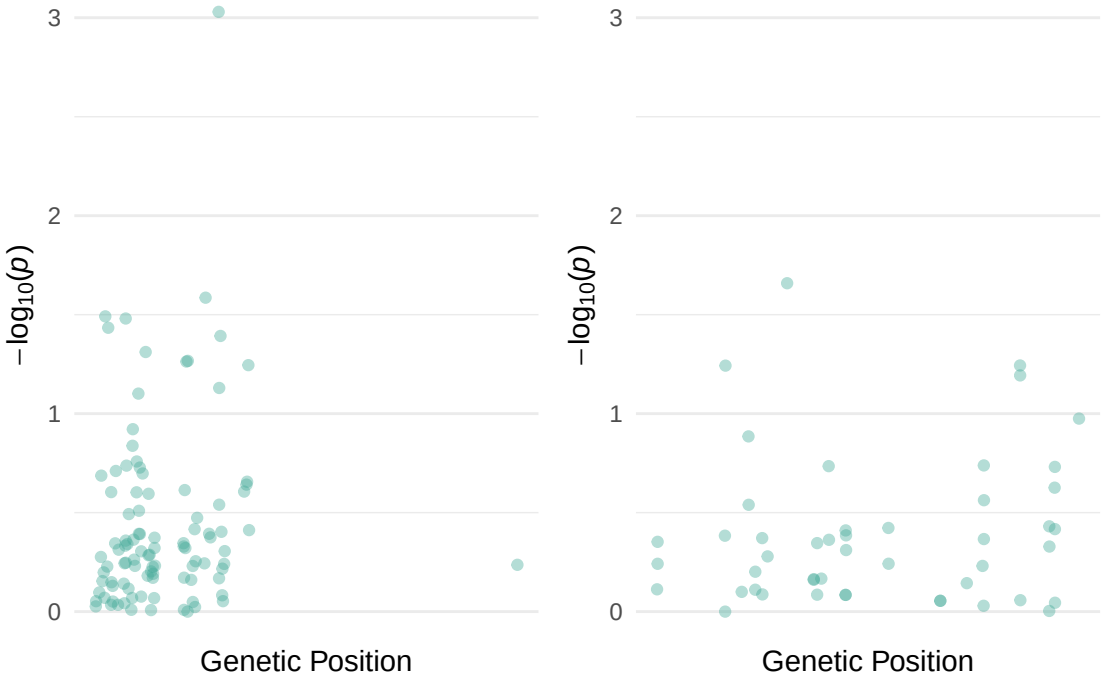


Figure 3.10: GWAS analyses at *PLPPR3* (left panel) and *COG3* (right panel) loci.

Chapter 4

Discussion and Outlook

The primary objective of this project was to identify parent-of-origin effects (POEs) within a gene-based burden testing framework, an approach designed to overcome the power limitations inherent in single-variant rare allele analysis. By disaggregating maternal and paternal haplotypes, we successfully identified significant POEs at several loci across the human exome. These signals were robustly detected across multiple functional categories—including missense and predicted loss-of-function (pLoF) variants—and persisted across a range of alternative allele frequency (AAF) bins. This confirms that parent-specific genetic architecture is not limited to a single class of variation but represents a pervasive layer of biological complexity that can be effectively captured through aggregated burden scores.

4.1 *KCNQ1*

A key step in our analysis was the replication of established parent-of-origin effects (POEs) to validate the integrity of our pipeline. A notable success was observed at the ***KCNQ1*** locus in relation to standing height, which has been previously identified as a POE candidate through SNP-based methodologies (Hofmeister *et al.*, 2025) but considering the goal of using these associations to validatethe intergrity of the pipeline, it shows that we have only moderate power to detect rare variant effect at known GWAS loci.

While previous research suggested that common variants inherited from the mother might be associated with a decrease in stature, our results—focused specifically on rare loss-of-function (LoF) variants—showed a potential divergence in directionality depending on the parental source.

In our analysis, the inheritance of a rare LoF variant on the maternal haplotype was associated with a tendency toward increased height, whereas paternal inheritance appeared to correlate with a decrease. This observation could potentially be explained by the epigenetic architecture of the *KCNQ1* cluster. The gene is typically expressed from the maternal allele, while the paternal allele is silenced by the antisense transcript ***KCNQ1OT1*** (Lawson *et al.*, 2013; Fitzpatrick *et al.*, 2002). If *KCNQ1* serves as a growth inhibitor, as suggested by its role in fetal development, the disruption of the functionally active maternal copy through an LoF mutation could theoretically lead to a loss of inhibition, potentially explaining the increase in phenotypic value.

However, the stronger negative association observed when these variants are inherited from the father suggests a more complex mechanism. It is possible that rare protein-altering variants on the paternal side do more than simply affect the silenced gene; they might interfere with the broader regulatory environment of the imprinted cluster, including the expression of the *KCNQ1OT1* silencer itself. These results highlight that the phenotypic impact of a variant may not only be contingent upon the parental origin but also upon the functional class of the variant being assessed. While our results align with the locus identified in Hofmeister *et al.* (2025),

the apparent "flip" in directionality suggests that the biological consequences of rare structural disruptions may differ significantly from those of common regulatory variants.

4.2 *SLC4A7*

The identification of a parent-of-origin effect at the *SLC4A7* locus represents a significant finding within our hypothesis-driven screen of candidate imprinted genes. *SLC4A7* (encoding the sodium-bicarbonate cotransporter NBCn1) has been previously characterized in the literature as exhibiting parent-of-origin (Zink *et al.*, 2018). This gene plays a fundamental role in intracellular alkalization, a process that is essential for cell growth, volume regulation, and metabolic homeostasis (Sedlyarov *et al.*, 2018).

In our analysis, we observed a bipolar association between LoF variants in *SLC4A7* and levels of lipoprotein A. Specifically, the inheritance of these variants on the maternal haplotype appeared to correlate with a decrease in lipoprotein A levels, whereas paternal inheritance of the same class of variants tended to associate with an increase.

This bipolar pattern might reflect an underlying epigenetic mechanism where the maternal and paternal alleles contribute differently to the regulation of lipid metabolism. If *SLC4A7* is involved in the clearance or synthesis pathways of lipoprotein A, a maternal LoF might disrupt a specific metabolic "investor" pathway, while a paternal LoF might interfere with a different regulatory branch of the same system. While these observations align with the gene's known biological functions in growth and metabolism, the limited number of carriers for these rare variants necessitates a cautious interpretation.

4.3 *NAPRT*

The identification of a POE at the *NAPRT* locus in association with Apolipoprotein B (ApoB) levels is consistent with its classification as an imprinted gene in clinical resources such as the Genomics England PanelApp. *NAPRT* encodes the enzyme nicotinate phosphoribosyltransferase, which is the rate-limiting step in the Preiss-Handler pathway, converting nicotinic acid (Vitamin B3) into Nicotinamide Adenine Dinucleotide (NAD+) (Covarrubias *et al.*, 2021).

Our results for missense variants in *NAPRT* reveal a potentially divergent phenotypic impact: maternal inheritance was associated with a decrease in ApoB levels, whereas paternal inheritance appeared to correlate with an increase. This observation is of particular interest given that nicotinic acid—the substrate for the *NAPRT* enzyme—is a known pharmacological agent used to lower ApoB and VLDL levels in clinical practice (Li *et al.*, 2019).

4.4 Exome-Wide Scan

Althought this project was initially aimed to assess and analyze rare variants, analyses were performed on common variants as well. Those represent the almost entire significant parent-of-origin associations identified in the genome-wide scan. The identified PofO signals were not part of the gene-trait pairs from strategy 1 neither from the list of imprinted genes suggesting new signals not picked up in previous GWAS studies, however, since we aggregated common variants into a burden score, the frequency of the burden can come close to 1, and this would not be a burden test anymore. Despite that, Figure 3.6 suggests strong signals in POE detection. The scan that filtered for rare variants only suggests poorer signal (Figure 3.8).

A major outcome of the exome-wide discovery scan (Strategy 3) was the identification of novel candidate loci showing robust evidence of parental asymmetry. Unlike the "bipolar" effects ob-

served in validation sets—where maternal and paternal variants move the phenotype in opposite directions—these signals were characterized by a pattern where the burden is driven almost entirely by one parental haplotype, while the other remains silent ($|\beta| \approx 0$).

4.4.1 *CPZ*

The identification of the *CPZ* (Carboxypeptidase Z) locus as a candidate for parent-of-origin effects shows consistent associations across a cluster of metabolic and liver-related traits. In our analysis, we observed a distinct maternal-specific influence on total cholesterol levels as well as on a highly related set of phenotypes, including LDL cholesterol, apolipoprotein B, alkaline phosphatase, and alanine aminotransferase. In all these instances, the phenotypic burden appeared to be driven by the maternal haplotype—with variants correlating with a tendency toward increased levels—while the paternal haplotype remained phenotypically silent. *CPZ* is known to be involved in the gallbladder and liver-related tissues (Deng *et al.*, 2025)

The detection of this maternal signal across multiple related biomarkers is can be expected given the physiological synergy between these traits. Apolipoprotein B is the primary structural component of LDL particles, which in turn constitute a significant portion of total circulating cholesterol (Kounatidis *et al.*, 2024). Furthermore, the presence of signals for alanine aminotransferase and alkaline phosphatase—established markers of hepatic function (Lala *et al.*, 2023)—suggests that the mechanism may be centered in the liver. As the liver is the primary organ responsible for lipoprotein assembly and cholesterol homeostasis, a gene that potentially modulates hepatic pathways or protein processing within liver cells would likely manifest its effects across this entire interconnected metabolic profile.

4.4.2 *AAK1*

The *AAK1* (AP2-associated kinase 1) locus exhibits a pattern of parental asymmetry similar to *CPZ*, where maternally inherited missense variants show a positive correlation with cholesterol-related phenotypes, while paternal inheritance remains phenotypically silent. However, a comparison of our results before and after the second quality control (QC2 required that any included in the burden score be observed at least once on both the maternal and paternal haplotypes across the cohort, see subsection 2.3.3) reveals a significant loss of statistical power for this signal (Figure 3.9). The decline in significance for AAK1 under QC2 suggests a potential lack of equivalent variant representation between the parental haplotypes. This implies that the initial signal may have been driven by "sampling sparsity"—where specific variants happened to be captured only on the maternal side in our dataset—rather than a definitive parent-of-origin biological mechanism. Consequently, while the maternal association is noteworthy, these findings show the importance of consistency filtering and suggest that the evidence for a POE at the *AAK1* locus should be interpreted with caution.

4.5 Suggestive Results

4.5.1 *NAPG* and *PLPPR3*

Beyond significant results, we also identified suggestive associations, such as *NAPG* and *PLPPR3*, associated with testosterone levels. Both genes exhibited a consistent bipolar association pattern, where maternally inherited LoF variants were associated with a tendency toward decreased testosterone levels, while the same class of variants on the paternal haplotype appeared to correlate with an increase.

The biological functions of these genes provide a plausible link to steroidogenesis. *NAPG* (N-ethylmaleimide-sensitive factor attachment protein gamma) is a central regulator of SNARE-

mediated vesicle trafficking (Bombardier and Munson, 2015). The synthesis of testosterone in Leydig cells is highly dependent on the mobilization of cholesterol into the mitochondria and the subsequent secretion of the hormone, both of which require efficient intracellular membrane trafficking and vesicle fusion pathways. A disruption in *NAPG* could theoretically impair the delivery of biosynthetic precursors or the export of the final hormone, leading to the observed shifts in systemic testosterone levels.

Similarly, *PLPPR3* (Phospholipid Phosphatase Related 3) is involved in modulating bioactive lipid signaling and the concentration of lipid phosphates at cellular membranes (Polyzou *et al.*, 2026). Since testosterone biosynthesis occurs within the specialized lipid environment of the smooth endoplasmic reticulum and the mitochondria, a gene that regulates the lipid microenvironment could reasonably be expected to influence the efficiency of the steroidogenic pathway.

The discovery of nearly identical bipolar phenotypic patterns across two distinct genomic regions suggests a potentially coordinated, parent-specific regulatory axis influencing testosterone homeostasis. This "tug-of-war" architecture is consistent with the predictions of the Parental Conflict Theory for hormonal signaling. However, as these loci have not been previously characterized as imprinted, these results should be interpreted as suggestive of a novel epigenetic layer of sex-hormone regulation that requires further molecular validation in steroidogenic tissues.)

4.6 Power

In the QQ plot for Strategy 1 (Figure 3.3), the observed $-\log_{10}(p)$ values fall consistently below the expected null distribution ($y < x$). This downward deviation suggests a notable lack of statistical power within this tier. Despite testing genes with established PofO associations, the restricted number of rare variant carriers within our cohort of 110,000 individuals likely prevents many true signals from reaching significance. The fact that the *KCNQ1* signal was nonetheless successfully identified confirms that while the strategy is underpowered for small effect sizes, it remains sensitive to loci with high biological impact.

Conversely, the QQ plots for the imprinted gene gcreen and the Exome-wide Discovery (Figures 3.5 and 3.6) show observed values that track closely with the null expectation before deviating upward at the tail, meaning that there is an enrichment of small $p$values compared to what is expected by chance. The adherence of the "body" of the distribution to the $y = x$ line indicates that the REGENIE model effectively controlled for genomic inflation, population stratification, and cryptic relatedness.

4.7 Limitations

The current study highlights that even with a substantial cohort of 110,000 individuals, the effective sample size for rare variants (MAF $\leq 1\%$) remains a limiting factor. On average, only approximately 1,000 individuals carry a variant of interest for any given gene. When these carriers must be further subdivided into maternal and paternal haplotypes to calculate a differential Z-score, the statistical requirements for robust discovery become extreme. Future analyses leveraging the full 500,000-participant UK Biobank release or meta-analyses across multiple biobanks will be essential to salvage the suggestive signals identified.

Chapter 5

Conclusion

This project successfully established a methodological framework for detecting parent-of-origin effects (POEs) using gene-based burden tests on phased rare-variant data. By separating maternal and paternal haplotypes, we demonstrated that the parental origin of a variant can be determinant and associated to a phenotype, revealing associations that are typically masked in standard additive genetic models.

Our global results confirm that parent-specific genetic architecture exists across a wide spectrum of complex traits. We identified two primary patterns of association: bipolar effects, where parental alleles influence a trait in opposite directions, and classical parent-of-origin effects, where one haplotype is silenced. These signals were notably concentrated within metabolic and hormonal phenotypes, providing empirical support for the role of genomic imprinting in regulating resource allocation and physiological homeostasis.

However, the study also highlights significant challenges in rare-variant discovery. The primary limitation remains statistical power; even within a large biobank cohort, the rarity of functional variants and the necessity for balanced haplotype representation restrict our ability to detect small effect sizes. This attenuation is anticipated: even in traditional additive models, the statistical signal from gene-based burden tests is generally lower in magnitude than that of a standard single-variant GWAS. Within a parent-of-origin context, signals are further diminished relative to additive frameworks. Consequently, the intersection of gene-level aggregation and parent-of-origin detection results in a compound reduction of overall statistical power. Furthermore, while our statistical evidence for parental bias is robust, these findings represent phenotypic associations that require future molecular validation—such as allelic expression or methylation studies—to definitively characterize the underlying epigenetic mechanisms.

Ultimately, this study shows that accounting for parental origin allows for the detection of associations beyond those captured by standard additive models. To build upon these findings, more work is required to enhance statistical power through larger cohorts, enabling a more exhaustive discovery of non-classical genetic influences

5.1 Reproducibility

Due to the sensitive nature and licensing restrictions of the UK Biobank Whole Exome Sequencing (WES) data, the raw genomic datasets are not available for download, which precludes the direct replication of the primary analysis outside of the host environment. All associations were conducted on the UK Biobank Research Analysis Platform (RAP), a secure, cloud-based ecosystem powered by DNAnexus that provides the high-performance computing resources necessary to process large-scale UKB data. While the full pipeline is dependent on this restricted data ac-

cess, example bash scripts demonstrating the implementation of the REGENIE burden tests have been provided to illustrate the analytical workflow. Furthermore, because of the significant computational overhead and memory requirements associated with exome-wide datasets, large-scale visualizations such as Manhattan plots were generated using cloud resources rather than local hardware. All scripts for data visualization and downstream processing are publicly accessible in the project's GitHub repository (https://gitlab.uzh.ch/martin.quintas/msc-thesis/-/tree/5889aeab871c6fa5102ffd690b437d15e583c494/).

Acknowledgement of AI Assistance

Parts of the coding and writing of this report were supported by the use of generative AI tools, specifically Google's Gemini (Google, 2026). This tool was used to assist with data manipulation in R, formatting LaTeX content, and improving the clarity and structure of written sections of the report. While all analyses and interpretations were performed and verified by the author, Gemini served as a helpful aid in enhancing workflow efficiency and refining the presentation of results.

Appendix A

Appendix

A.1 Supplementary Figures

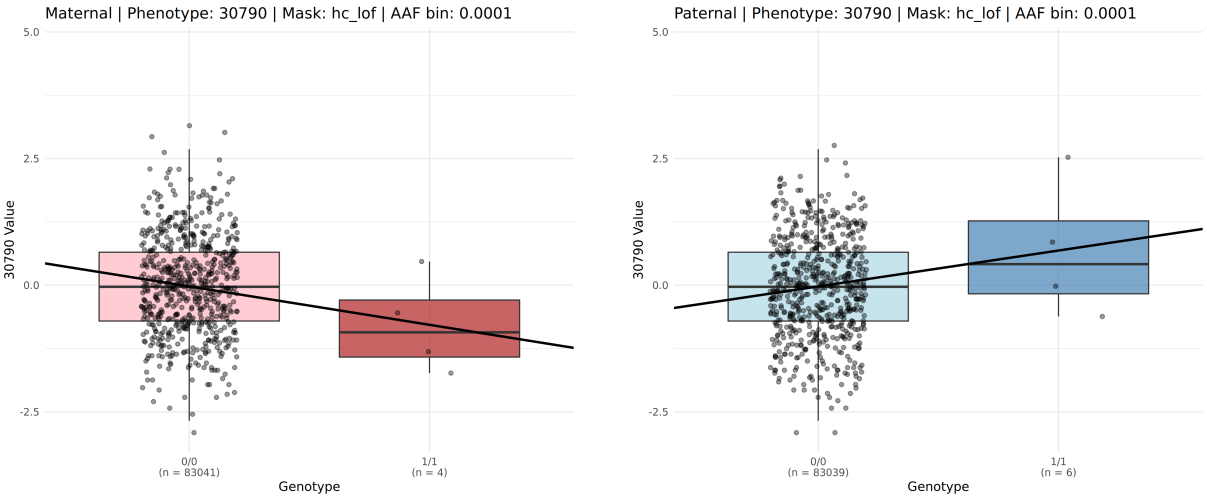


Figure A.1: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized lipoprotein A (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *SLC4A7* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

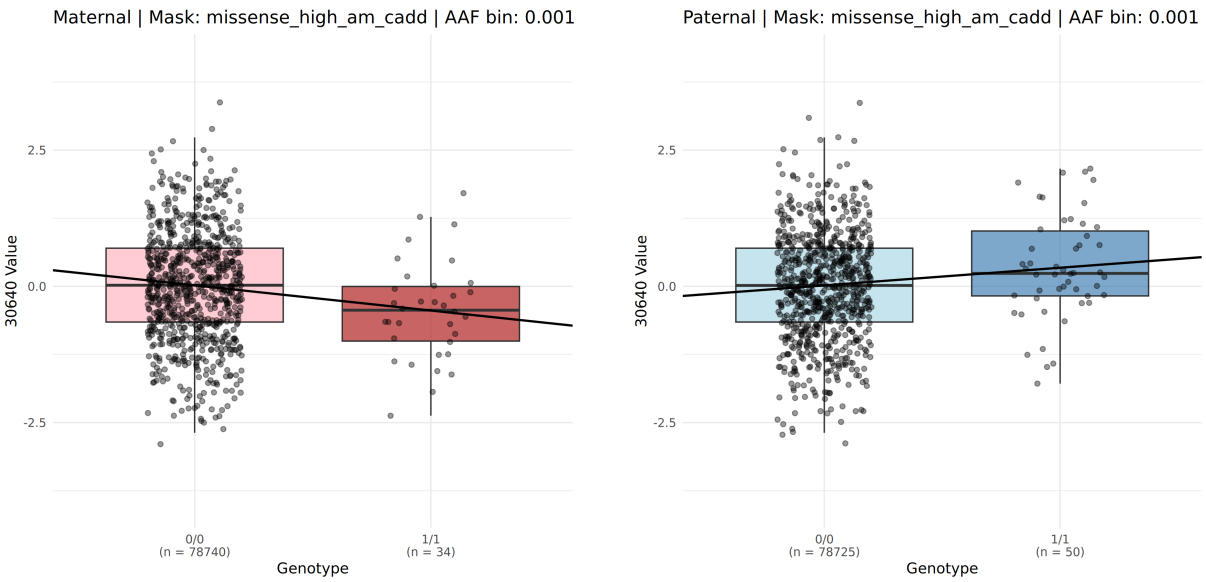


Figure A.2: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized apolipoprotein B (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *NAPRT* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

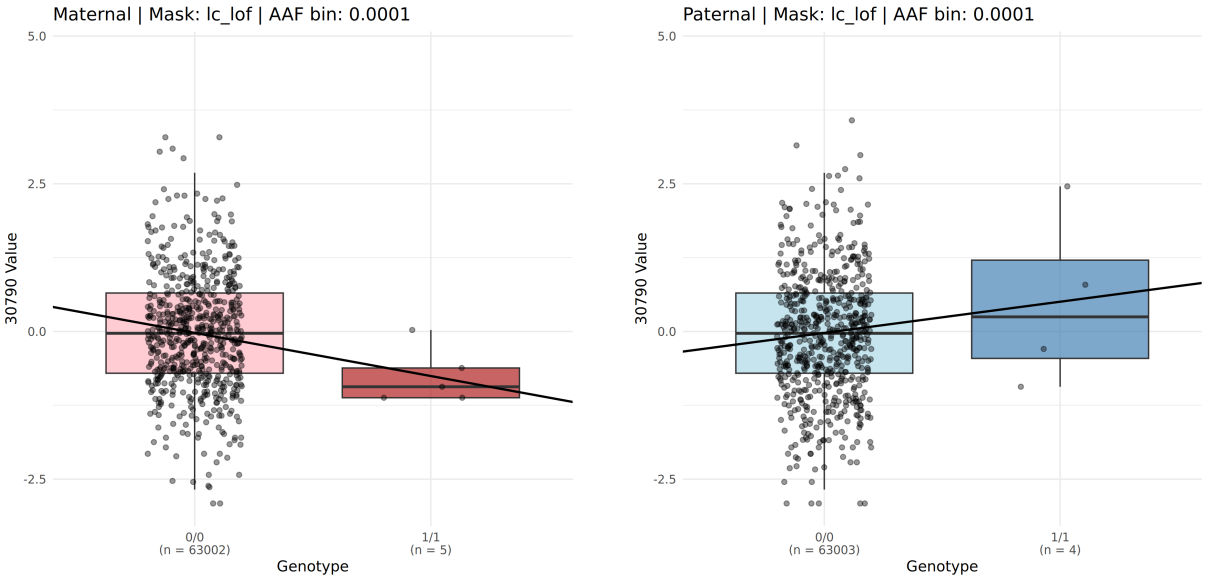


Figure A.3: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized lipoprotein A (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *CST1* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

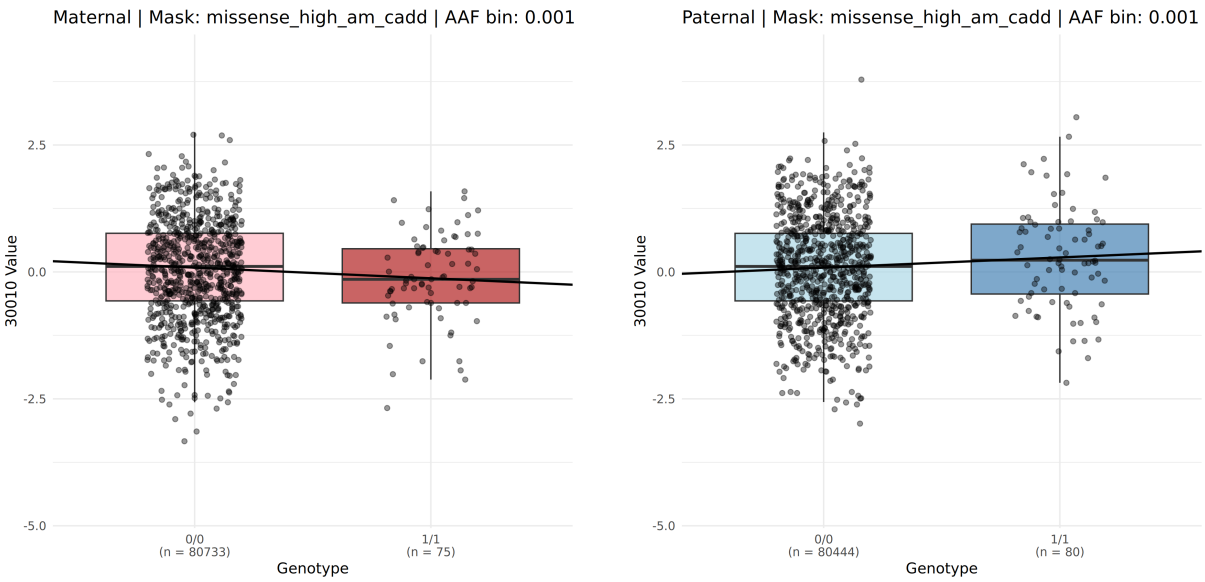


Figure A.4: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized erythrocyte count (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *PLG* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

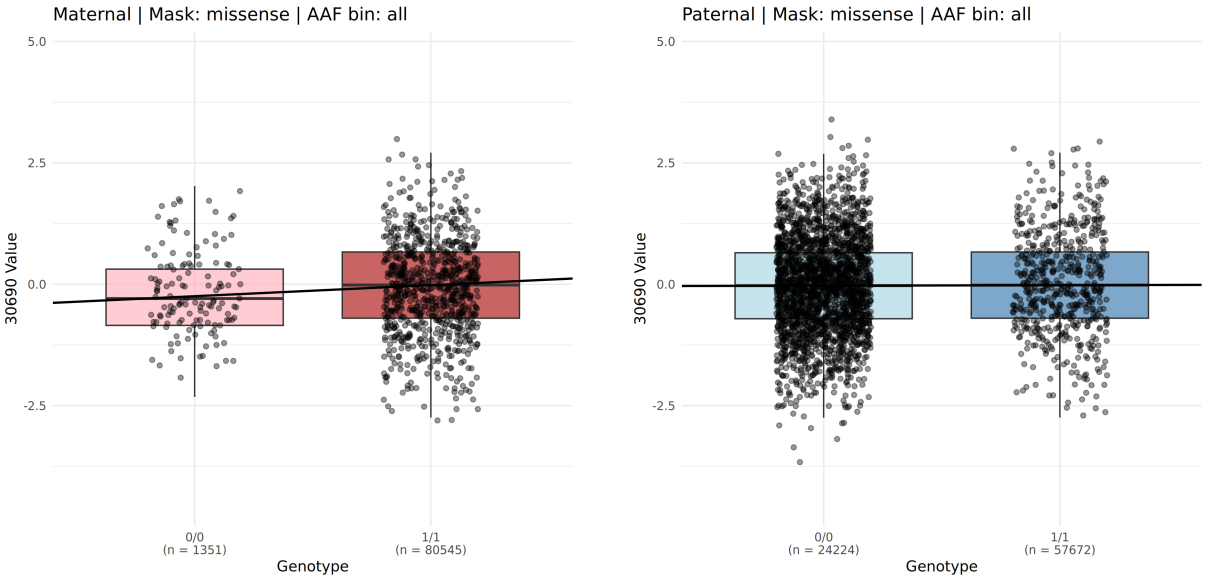


Figure A.5: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized cholesterol (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *CPZ* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

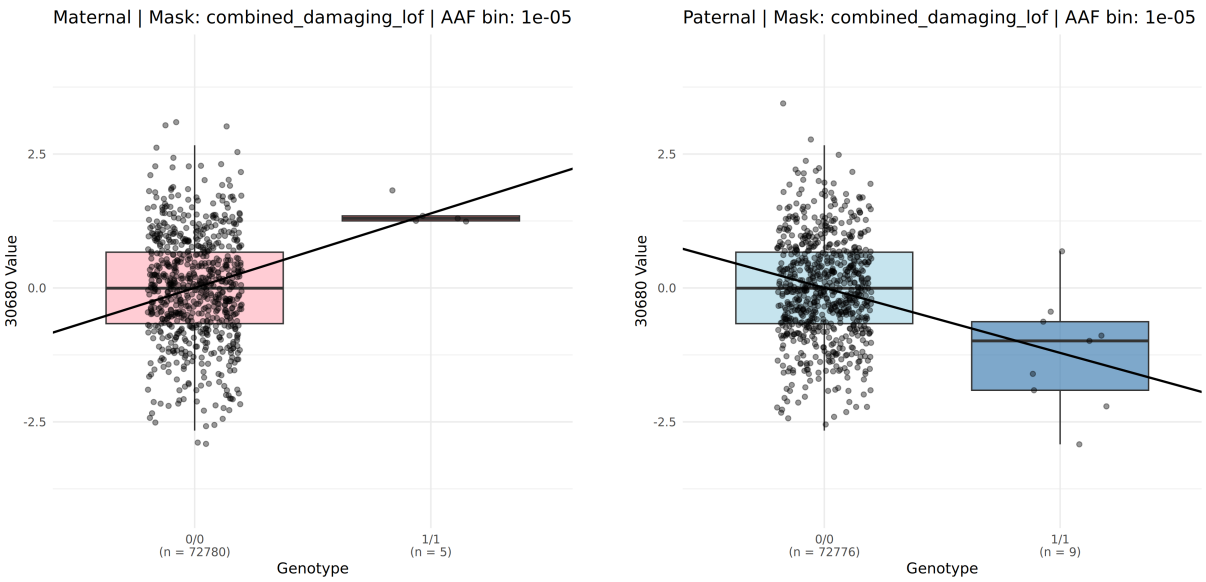


Figure A.6: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized calcium level (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *FYB1* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

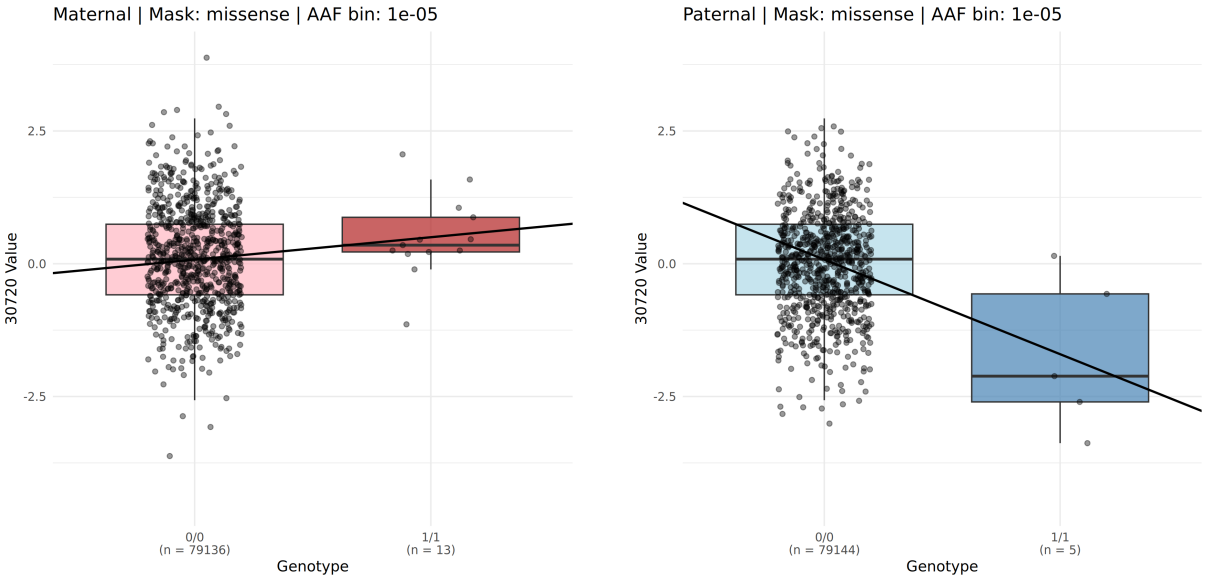


Figure A.7: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized cystatin C level (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *CTSG* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

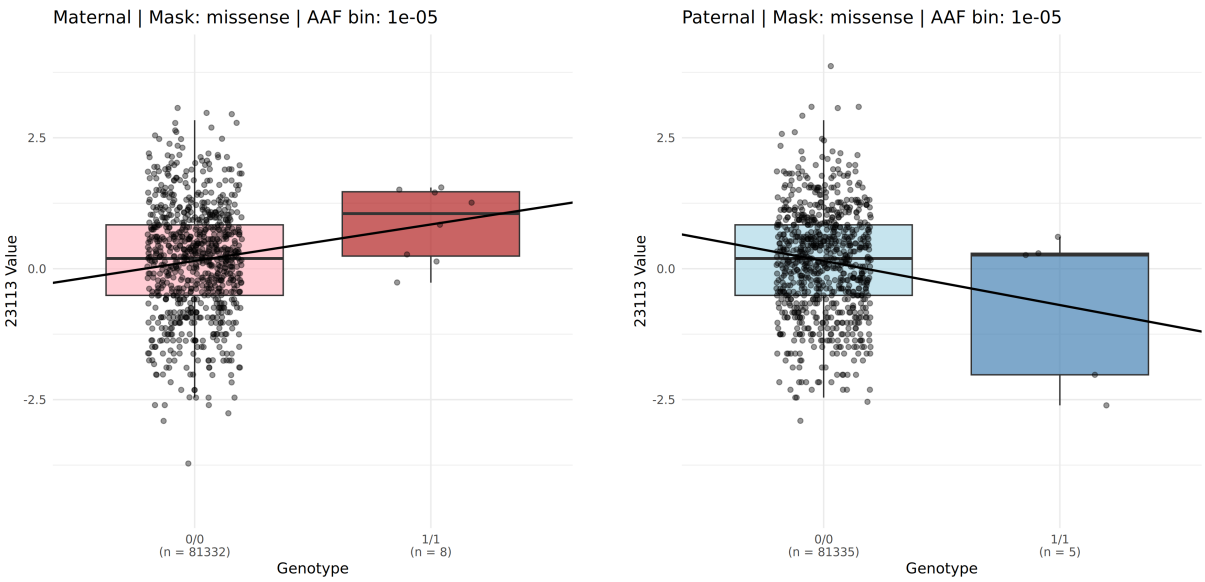


Figure A.8: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized leg fat-free mass (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *THAP10* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

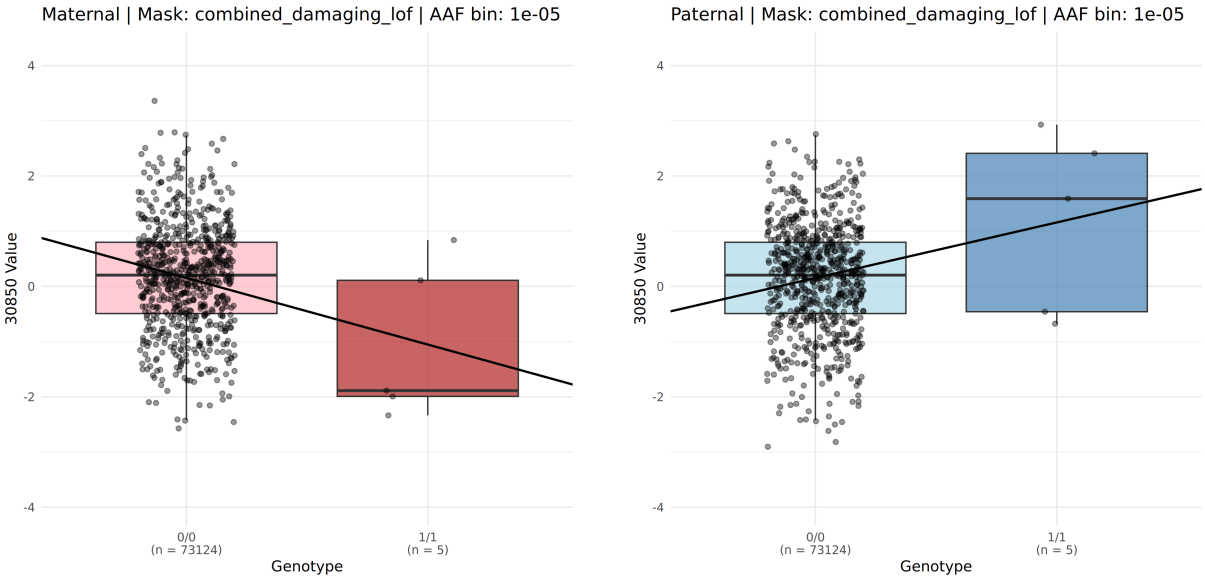


Figure A.9: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized testosterone level (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *NAPG* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

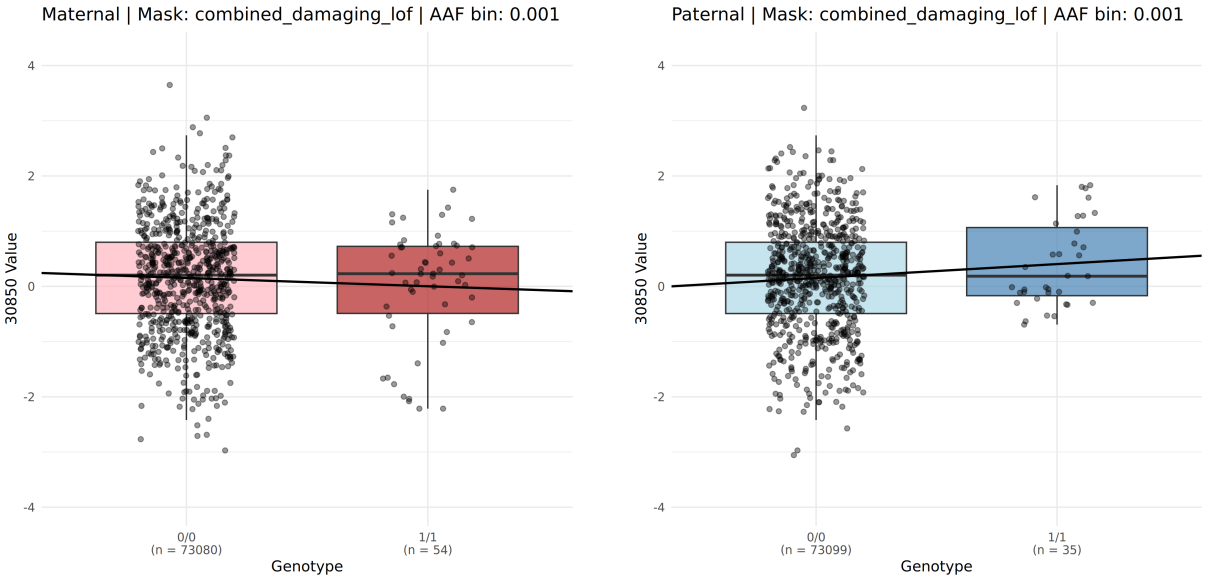


Figure A.10: Presence or absence of rare variants (0/0 and 1/1, respectively, x-axis) effect on normalized testosterone level (y-axis), for the maternal (red, left) and paternal (blue, right) haplotype at *PLPPR3* locus. Each dot represent an individual. Boxes indicate the interquartile range (IQR), with the bottom and top of the box representing the 25th (Q1) and 75th (Q3) percentiles, respectively. The horizontal line within the box represents the median (50th percentile). Whiskers extend to the smallest and largest values within Q1$-1.5 \times$ IQR and Q3 $+ 1.5 \times$ IQR.

A.2 Supplementary Information

A.2.1 Computational Details

This document was generated on March 13, 2026 at 22:17. R version and packages used to generate this report:

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## Running under: macOS Tahoe 26.3.1
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##
## loaded via a namespace (and not attached):
##  [1] vctrs_0.6.5        cli_3.6.5          rlang_1.1.6        xfun_0.54
##  [5] highr_0.11         generics_0.1.4     S7_0.2.1           glue_1.8.0
##  [9] scales_1.4.0       grid_4.5.2         evaluate_1.0.5     tibble_3.3.0
## [13] lifecycle_1.0.4    compiler_4.5.2     codetools_0.2-20   RColorBrewer_1.1-3
## [17] Rcpp_1.1.1         pkgconfig_2.0.3    rstudioapi_0.17.1  farver_2.1.2
## [21] digest_0.6.37      R6_2.6.1           tidyselect_1.2.1   pillar_1.11.1
## [25] magrittr_2.0.4     tools_4.5.2        withr_3.0.2        gtable_0.3.6
```

A.2.2 Regenie Step 1

```
#!/bin/bash

wget https://github.com/rgcgithub/regenie/releases/download/v4.1/regenie_v4.1.gz_x86_64_Linux.z
unzip regenie_v4.1.gz_x86_64_Linux.zip
chmod +x ./regenie_v4.1.gz_x86_64_Linux

# Input files
BED=my_genotype_array_data   # UK Biobank SNP array genotype data (bed format)
SNP=my_qc_passed_snplist.txt  # list of QC-passed SNPs for step 1
```

```
SMP=my_qc_passed_samples.txt  # list of QC-passed sample IDs
PHF=my_phenotype_file.txt.gz  # phenotype file for the target trait
COF=my_covariates.txt.gz    # covariate file (age, sex, PCs)

# Run REGENIE step 1 (whole-genome regression)
./regenie_v4.1.gz_x86_64_Linux --step 1 \
  --bed ${BED} \
  --extract ${SNP} \
  --keep ${SMP} \
  --phenoFile ${PHF} \
  --covarFile ${COF} \
  --covarColList PC{1:20},age,sex \
  --qt \
  --bsize 1000 \
  --lowmem \
  --lowmem-prefix regenie_tmp_preds \
  --threads 4 \
  --out my_step1_output
```

A.2.3 Regenie Step 2

```
#!/bin/bash

# Dependencies
wget https://github.com/rgcgithub/regenie/releases/download/v4.1/regenie_v4.1.gz_x86_64_Lin
unzip regenie_v4.1.gz_x86_64_Linux.zip
chmod +x ./regenie_v4.1.gz_x86_64_Linux

# Input files
CAU=my_selected_individuals.txt     # file listing selected individual IDs
COF=my_covariates.txt.gz            # covariate file (age, sex, PCs)
PHF=my_phenotype_matrix.txt         # phenotype matrix
ANO=my_annotations.txt              # variant functional annotations
SET=my_setlist.txt                  # gene set list for burden grouping
MSK=my_mask_definitions.txt         # mask definitions for variant aggregation
AAF=my_allele_frequencies.txt       # ancestry-specific allele frequency file
LST=my_step1_predictions.txt        # REGENIE step 1 prediction list

# Run REGENIE step 2 (association testing)
./regenie_v4.1.gz_x86_64_Linux --step 2 \
    --pgen my_haplotype_data \       # input genotype file (pgen format)
    --phenoFile ${PHF} \
    --covarFile ${COF} \
    --covarColList PC{1:20},age,sex \
    --pred ${LST} \
    --keep ${CAU} \
    --anno-file ${ANO} \
    --set-list ${SET} \
    --mask-def ${MSK} \
```

```
    --aaf-bins 1,0.1,0.01,0.001,0.0001,0.00001 \
    --aaf-file ${AAF} \
    --build-mask max \
    --vc-tests acatv \
    --joint acat \
    --vc-MACthr 20 \
    --approx --pThresh 0.05 \
    --bsize 400 \
    --threads 8 \
    --out my_output_prefix
```

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